Brain Tumour Detection and Classification Using Deep Learning: A Comprehensive Comparative Study of CNN and Transformer Architectures

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Abstract

The presence of brain tumours can be considered one of the most complicated neurological problems in the world, where early and precise diagnosis can determine the success of therapy and ensure survival. Magnetic resonance imaging (MRI) is the standard procedure for diagnosis, yet manual image interpretation takes much time, is subject to inter-observer variability, and needs experts whose competence may not always be at hand. In this paper we provide a systematic comparison of nine deep learning models for brain tumour detection and for classification into glioma, meningioma, and pituitary tumours. Classical convolutional neural networks (ResNet50, VGG16, DenseNet121), efficient convolutional neural networks (EfficientNetB0, MobileNetV3-Large, InceptionV3), and transformer or hybrid transformer models (ViT-B/16, Swin-Base, CoAtNet-0) are evaluated under a single standardised protocol. Perceptual hashing applied before splitting removed 19.3% near-duplicate images from the multiclass collection, reducing 5,828 MRI images to 4,705 unique samples; binary detection used a separate set of 3,253 images. All models were initialised from ImageNet weights and fine-tuned using the AdamW optimiser, a cosine annealing learning rate schedule, and mixed precision training on two NVIDIA Tesla T4 graphics processing units. On the held-out test set, CoAtNet-0, DenseNet121, EfficientNetB0, and ResNet50 each reached 99.39% binary detection accuracy, with DenseNet121 and ResNet50 attaining an area under the receiver operating characteristic curve (AUC) of 1.0000. For multiclass classification, Swin-Base achieved the highest accuracy of 98.58% with a macro-averaged one-versus-rest AUC of 0.9993, followed by CoAtNet-0 at 98.44% and InceptionV3 at 98.30%. EfficientNetB0 offered the best accuracy-to-cost balance, reaching 99.39% detection accuracy and 97.59% subtype accuracy with only 4.0 million parameters, 0.77 GFLOPs per image, and 10.8 ms single-image latency. Per-class sensitivity, specificity, and F1 scores, calibration error, gradient-weighted class activation mapping, and measured inference cost are reported for every model.

Keywords: Brain tumour detection, deep learning, convolutional neural networks, vision transformers, medical image classification, MRI, transfer learning

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1. Introduction

Brain tumours constitute the abnormal proliferation of cells either in the brain or the central nervous system. Whether benign or malignant, brain tumours represent serious threats because of their location in the limited skull cavity and the important functions of the brain. According to the American Brain Tumor Association, about 90,000 people are estimated to have primary brain tumours each year in the USA alone, while the global scenario is similar with respect to incidence. The exact prevalence in India, however, is not fully documented. Nonetheless, in the major metropolitan cities such as Mumbai, Delhi, Bangalore, and Chennai, the tertiary care hospitals report thousands of patients with primary brain tumours each year. Three major forms of primary brain tumours include: (i) glioma, which is formed from glial cells and constitutes 30% of total brain tumours; (ii) meningioma, which develops in the meningeal layers of the brain and is usually benign in nature; and (iii) pituitary tumours, which grow in the pituitary gland and cause hormonal problems [1, 2].

Magnetic resonance imaging (MRI) is used nowadays as the main non-invasive technique for brain tumour detection and classification. In comparison to traditional computed tomography imaging, MRI gives much better soft tissue contrast without utilising any ionising radiation. Moreover, MRI allows images to be acquired in different orientations and with different types of contrast, namely T1-weighted, T2-weighted, fluid-attenuated inversion recovery (FLAIR), and diffusion-weighted. This is extremely important for determining tumour location, volume, margins, oedema, and adjacent vital structures. At the same time, analysis of MRI studies raises a number of problems that cannot be ignored. Firstly, examination takes a lot of time; a complete brain MRI study may include hundreds of sections, and it takes a specialist radiologist 15–30 minutes to make a correct diagnosis [16, 17]. Secondly, inter- and intra-observer variability should be noted, as there is no guarantee that radiologists will interpret the results of an MRI study in the same way. Thirdly, limited training in neuroradiology can become an obstacle to correct image interpretation. Finally, there are not enough specialists in the rural and semi-urban regions of India because of the poor radiologist-to-population ratio.

The introduction of deep learning technology has transformed the field of medical imaging analysis in the last ten years. Models based on deep learning, especially convolutional neural networks (CNNs), have the ability to learn hierarchical representations of visual features directly from pixel inputs without the need for manual feature extraction. Thus, the use of these models enables automatic, efficient, and consistent tumour detection from brain MRI images. Convolutional neural networks have become popular among researchers working in this area, with well-performing models such as ResNet, VGG, DenseNet, Inception, EfficientNet, and MobileNet setting benchmarks on image classification tasks. More recently, researchers started exploring transformer models for computer vision problems, having been introduced initially for natural language processing. These transformer-based models include the vision transformer (ViT), the Swin Transformer, and CoAtNet. This line of research provides an alternative way of representing global information in images using the self-attention mechanism, and it might be of particular use in medical imaging, especially for understanding the relation between a tumour and the surrounding anatomical structures [22, 23, 24, 25].

Although several research works have separately proved the efficiency of deep learning architectures for detecting and classifying brain tumours, there is still an insufficient number of comparative investigations performed across several architectural families using the same experimental settings. The lack of comparative analysis poses serious challenges in choosing a suitable model, since each study employs a separate database, preprocessing procedure, data augmentation approach, and assessment technique, which makes it problematic to compare results reported by different studies. For example, it is impossible to say which performance figure is better when one work reports an accuracy of 99% on a small and homogeneous database, whereas another gives 97% accuracy on a larger and heterogeneous one.

Numerous papers make the mistake of retaining images that are duplicates or near-duplicates across the training, validation, and test splits when using publicly available datasets drawn from multiple sources. This results in a highly inaccurate assessment of performance, causing a significant increase in reported performance figures compared with the actual ability of a model to generalise. Once a model has been shown a particular image, or one that is very similar, during training, any further performance result for that image reflects memorisation rather than genuine learning. The problem of near-duplicate images is especially critical within brain tumour research, since most publicly available datasets come from common origins such as the Cheng dataset or the Figshare collection. Most papers that address this problem use simple exact-matching techniques that rely either on the file name or on the cryptographic hash of an image. However, exact-matching techniques fail to detect even simple modifications such as changing the size of an image, recompressing it, rotating it slightly, or altering its contrast.

It is important to emphasise the clinical significance of establishing a brain tumour diagnosis correctly. The potential consequences of late diagnosis or misdiagnosis include limited treatment alternatives, poor prognosis, higher healthcare expenditure, and significantly diminished quality of life for both patients and their families. The implementation of automated systems with accuracy equal or superior to that of humans may bring about several advantages: faster diagnosis through prompt preliminary assessment; decision-making support where experienced radiologists are unavailable; uniformity of diagnosis due to the absence of inter-reader variation; and relief of radiologist workload so that experts attend primarily to challenging cases [10, 11].

The purpose of this paper is to achieve several objectives. Firstly, the implementation of nine deep learning models ranging from classical convolutional neural networks (ResNet50, VGG16, DenseNet121) through efficient networks (EfficientNetB0, MobileNetV3-Large, InceptionV3) to transformer and hybrid transformer models (ViT-B/16, Swin-Base, CoAtNet-0). Secondly, the establishment of a common testing protocol based on identical train, validation, and test splits, identical preprocessing including perceptual hashing for image deduplication, and comparable hyperparameters across all models. Thirdly, the assessment of model performance by means of clinically meaningful metrics including accuracy, per-class sensitivity and specificity, F1 score, area under the receiver operating characteristic curve, calibration error, and efficiency metrics such as parameter count, floating point operations, training time, and measured inference latency. Fourthly, the exploration of architectural trends in brain tumour analysis in order to determine whether residual, dense, multi-scale, attention-based designs, or neural architecture search improve performance. Fifthly, the provision of evidence-based recommendations on model selection for different deployment scenarios based on the trade-off between accuracy and efficiency.

The rest of this paper is structured as follows. Section 2 reviews existing studies on brain tumour classification. Section 3 describes the complete methodology adopted in this study, including the overall workflow, data collection, preprocessing, model development, and experimental settings. Section 4 presents and discusses the results of both the binary detection and the multiclass classification tasks, together with efficiency and interpretability analyses. Section 5 concludes the study and offers directions for future research.

2. Literature Review

Previous works on automated brain tumour recognition were based on feature engineering followed by machine learning classifiers. Such works have used texture descriptors such as the grey level co-occurrence matrix (GLCM) and local binary patterns (LBP), wavelet transforms such as the discrete wavelet transform (DWT), morphological features related to shape, size, and boundary, and classifiers such as the support vector machine (SVM), random forests, and *k*-nearest neighbours [2, 3]. In their work, Zacharaki et al. [1] obtained around 85% accuracy using texture features and an SVM classifier on a small set of 80 images for tumour type recognition. Cheng et al. [5] suggested a tumour region augmentation approach that yielded 90% accuracy using intensity, texture, and shape features along with an SVM classifier on 3,064 MRI images. Although promising, such methods faced challenges due to the limitations of feature engineering, the inability to generalise across datasets, and the absence of automatic learning of hierarchical structures.

The success of AlexNet in 2012 propelled the implementation of deep learning in medical imaging applications. Successive CNN models improved performance by introducing better architectures. He et al. [6] incorporated residual connections into CNN models, making it possible to train very deep networks by addressing the vanishing gradient problem. ResNet50 became a standard model for medical image classification, with reports of 95–98% accuracy for brain tumour classification. Using transfer learning on a ResNet50 model, Seetha and Raja [9] achieved an accuracy of 97.5% for the classification of 253 brain MRI images. Simonyan and Zisserman [4] demonstrated improvements in model performance by increasing depth while keeping small convolutional filters of size 3×3. However, due to its very large parameter count of 138 million, VGG16 is computationally intensive and vulnerable to overfitting when applied to small medical datasets. For instance, Khan et al. [18] reported 98.69% accuracy using VGG16 with 3,064 samples. Huang et al. [8] introduced densely connected layers in which each layer receives input from all preceding layers. Consequently, DenseNet121 became more parameter-efficient while achieving better performance in medical imaging tasks, with accuracies of 96–98%. Szegedy et al. [7] presented parallel convolutions with varying kernel dimensions (1×1, 3×3, 5×5) that allowed the model to learn multi-scale features. InceptionV3 reached 97.5% accuracy in brain tumour classification, as reported by Deepak and Ameer [12]. Moreover, Rehman et al. [20] obtained a high classification rate of 99.34% accuracy via transfer learning. Tan and Le [14] introduced a method for scaling network dimensions, namely depth, width, and resolution, using neural architecture search to obtain better accuracy with far fewer parameters than comparable models. Furthermore, Howard et al. [15] created lightweight neural networks based on depth-wise separable convolutions for mobile and edge devices.

Dosovitskiy et al. [22] adapted transformer models to visual tasks by framing images as sequences of patches of 16×16 pixels. The ViT-B/16 variant performed comparably with CNNs on large-scale datasets such as ImageNet and JFT-300M but needed extensive pre-training and exhibited drawbacks on small medical datasets because of a lack of built-in inductive biases. Liu et al. [23] proposed hierarchical transformers using shifted windows to process images efficiently at multiple scales while retaining linear complexity with respect to image resolution. Swin Transformers performed impressively on ImageNet and have also been investigated for medical image processing. The hierarchical window attention in Swin Transformers enables the model to learn both local tumour boundaries and overall structure from MRI images. CoAtNet, proposed by Xu et al. [24], employs convolution and attention simultaneously within a single model, which is intended to combine the inductive bias of convolution with the global receptive field of attention.

Recent studies applying these architectures to brain tumour detection include the comparative summary presented in Table 1.

Table 1: Comparison of existing studies for brain tumour detection.

Study	Model	Dataset size	Task	Accuracy
Talo et al. (2019) [13]	ResNet50	253	Binary	95.23%
Khan et al. (2020) [18]	VGG16	3,064	Binary	98.69%
Noreen et al. (2020) [19]	DenseNet121	3,064	Binary	99.51%
Rehman et al. (2020) [20]	InceptionV3	3,064	Binary	99.34%
Ghassemi et al. (2020) [21]	EfficientNet	3,064	Binary	98.87%
Cheng et al. (2015) [5]	SVM + features	3,064	Multiclass	90.00%
Deepak and Ameer (2019) [12]	InceptionV3	3,064	Multiclass	97.50%

Several open issues remain. Cross-architecture comparison is difficult because most studies examine no more than two or three models. Dataset heterogeneity is a further obstacle, because split schemes, preprocessing, and evaluation protocols differ between studies. Transformer under-representation is an additional problem, since current papers deal mainly with CNN-based models. Reproducibility is also limited, because source code, exact splits, and pretrained weights are not consistently released. Incomplete deduplication is a final gap, as most papers do not account for duplicate and near-duplicate images in combined datasets. Reporting practice is also narrow: headline accuracy and AUC are commonly quoted without per-class sensitivity and specificity, without any assessment of probability calibration, and without measured inference cost, all of which matter for clinical interpretation. This paper addresses these gaps by evaluating nine architectures under identical conditions, combining datasets from multiple public sources, applying perceptual hash based deduplication before splitting, comparing CNN and transformer families directly, reporting per-class and calibration metrics alongside measured latency and throughput, and releasing the complete code and per-model artefacts for both the binary and the multiclass tasks.

3. Methods and Materials

3.1. Overall Methodology

The complete research workflow adopted in this study is illustrated in Figure 1. The pipeline proceeds through five stages. In the data acquisition stage, two publicly available Kaggle collections are combined for the binary detection task and two further collections are combined for the multiclass subtype task. In the data curation stage, perceptual hashing is applied to the multiclass collection to remove near-duplicate images before any split is drawn, after which both datasets are partitioned into training, validation, and test subsets using stratified sampling so that class proportions are preserved in every subset. In the preprocessing stage, images are converted to three-channel form, resized to the input resolution expected by each backbone, normalised using ImageNet statistics, and augmented on the fly during training only. In the model development stage, nine ImageNet-pretrained backbones are fine-tuned under a single shared training protocol, with the classification head of each backbone replaced to match the number of target classes. In the evaluation stage, the checkpoint selected on the validation split is applied once to the held-out test split, and accuracy, per-class sensitivity and specificity, F1 score, AUC, calibration error, gradient-weighted class activation maps, and inference cost are recorded for every model.

A design point worth stating explicitly is that the two tasks are kept separate rather than merged into a single four-class problem. The binary detector answers whether a tumour-like abnormality is present, whereas subtype classification is only meaningful once a tumour-positive decision has already been taken. Keeping the tasks separate mirrors the clinical order of interpretation and avoids making the subtype model responsible for rejecting normal images that were never part of its training objective.

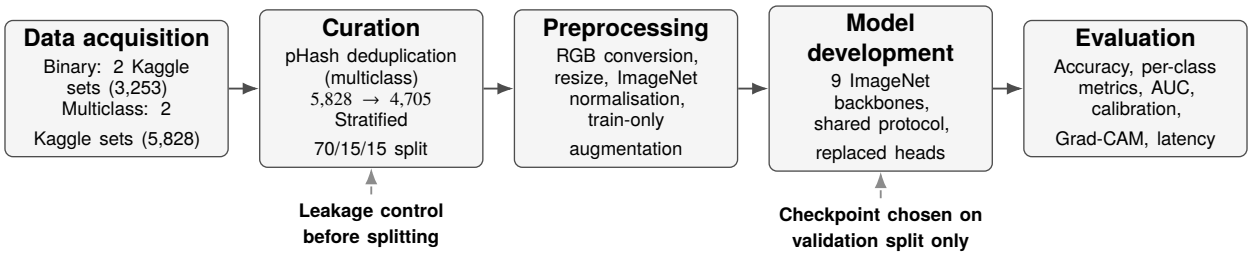


Figure 1: Overall methodology of the study, showing the five stages of the pipeline from data acquisition through curation, preprocessing, and model development to evaluation. Perceptual hash deduplication is applied before splitting so that near-duplicate images cannot leak between training and test subsets, and the test split is used only once, with the checkpoint selected on the validation split.

3.2. Data Collection

For the binary brain tumour detection task, which requires classifying MRI images as either containing a tumour or being tumour-free, we combined two publicly available brain MRI datasets sourced from the Kaggle platform. Dataset 1, *Brain MRI Images for Brain Tumor Detection* [26], contains 253 MRI images in total, with 155 labelled as tumour-positive and 98 as tumour-negative. The images represent a mix of T1-weighted and T2-weighted sequences, though specific sequence information is not uniformly provided. Dataset 2, *Br35H: Brain Tumor Detection 2020* [27], contains a substantially larger collection of 3,000 MRI images with a perfectly balanced distribution of 1,500 tumour-positive and 1,500 tumour-negative images. The images are provided in JPEG format and have been pre-cropped to focus on the brain region, though some variability in field of view and image resolution exists across samples. The combined dataset for binary classification comprised 3,253 images, with 1,598 tumour-negative images (49.1%) and 1,655 tumour-positive images (50.9%). This near-perfect balance prevents the model from exploiting class imbalance and keeps chance-level accuracy at approximately 50%. Representative samples of tumour-negative and tumour-positive images are shown in Figure 2, and the class distribution is given in Figure 3.

For the more challenging multiclass classification task, in which the model must distinguish between glioma, meningioma, and pituitary tumours, we combined two further publicly available brain tumour MRI datasets. Dataset 1, Bhuvaji et al., *Brain Tumor Classification (MRI)* [28], provides all images in JPEG format at a consistent resolution of 224×224 pixels. Dataset 2 is the brain tumour collection of Cheng et al. [29], described in the associated study [5], and was obtained in the JPEG conversion distributed on Kaggle [30]. The combined dataset prior to deduplication consisted of 5,828 images, with glioma the largest subset at 2,352 images (40.4%), meningioma at 1,645 images (28.2%), and pituitary at 1,831 images (31.4%). Representative samples of each tumour type are shown in Figure 4.

The two collections differ substantially in their native image geometry, as summarised in Table 2 and illustrated in Figures A.2 and B.3. The binary detection images are heterogeneous, with widths ranging from 150 to 1,920 pixels and a standard deviation of 155 to 195 pixels about mean dimensions of roughly 300 to 400 pixels, reflecting the varied cropping and field of view of the two contributing sources. The multiclass images are far more uniform, with a median of 512×512 pixels in every class and interquartile ranges that collapse to a single value, since the great majority of images were distributed at that resolution. This difference matters for interpretation: the multiclass task operates on a visually homogeneous collection, whereas the binary task requires the model to absorb considerably more variation in scale and framing. All images were resized to a common input resolution during preprocessing, so this heterogeneity affects the effective magnification of anatomy rather than the tensor shape presented to the network.

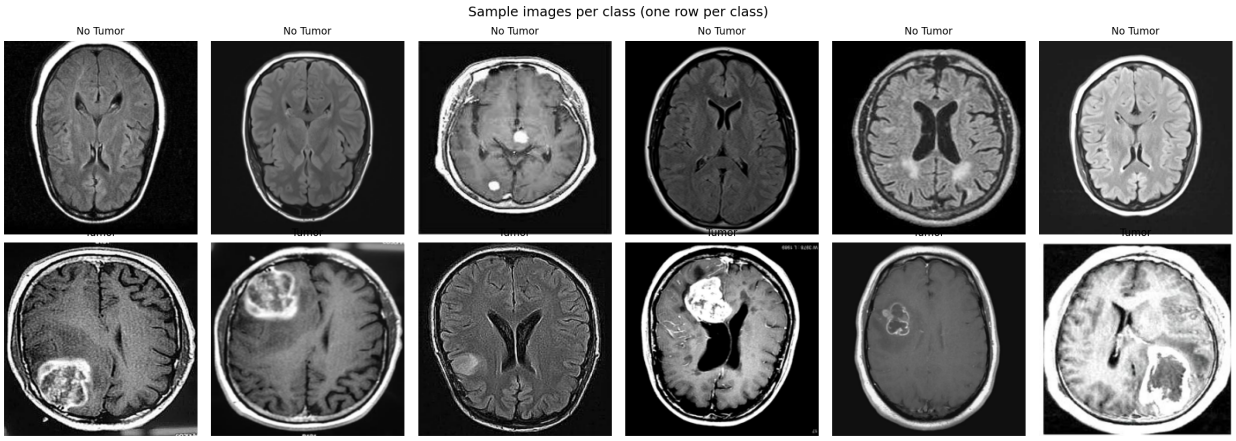


Figure 2: Sample brain MRI images used for the binary detection task. The top row shows tumour-negative images and the bottom row shows tumour-positive images.

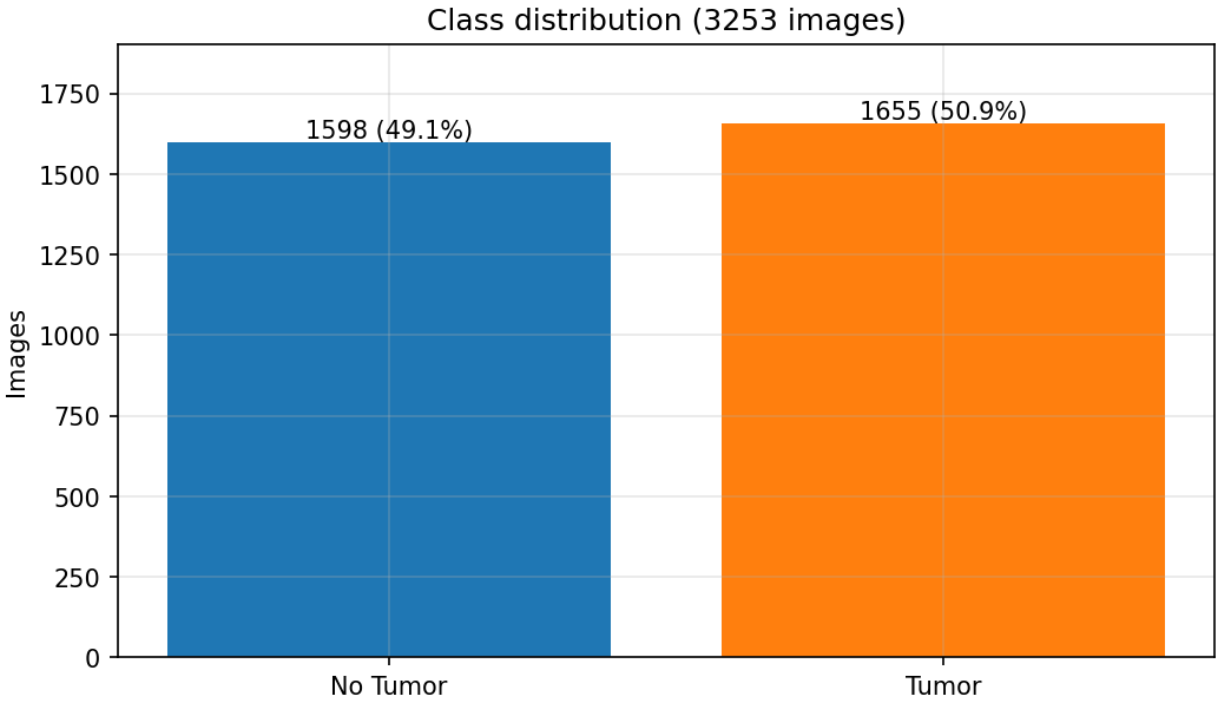


Figure 3: Class distribution of the combined binary detection dataset, comprising 1,598 tumour-negative and 1,655 tumour-positive images.

Table 2: Native image dimensions of the source collections before resizing, in pixels. Values are computed across all images of each class prior to deduplication.

Task	Class	Images	Mean W	Mean H	SD W	Min W	Max W
Binary	No tumour	1,598	302.6	308.9	154.9	150	1,920
	Tumour	1,655	351.5	400.2	195.2	175	1,275
Multiclass	Glioma	2,352	507.6	508.4	33.0	175	640
	Meningioma	1,645	483.1	485.9	90.4	174	1,275
	Pituitary	1,831	512.5	512.2	84.4	201	1,365

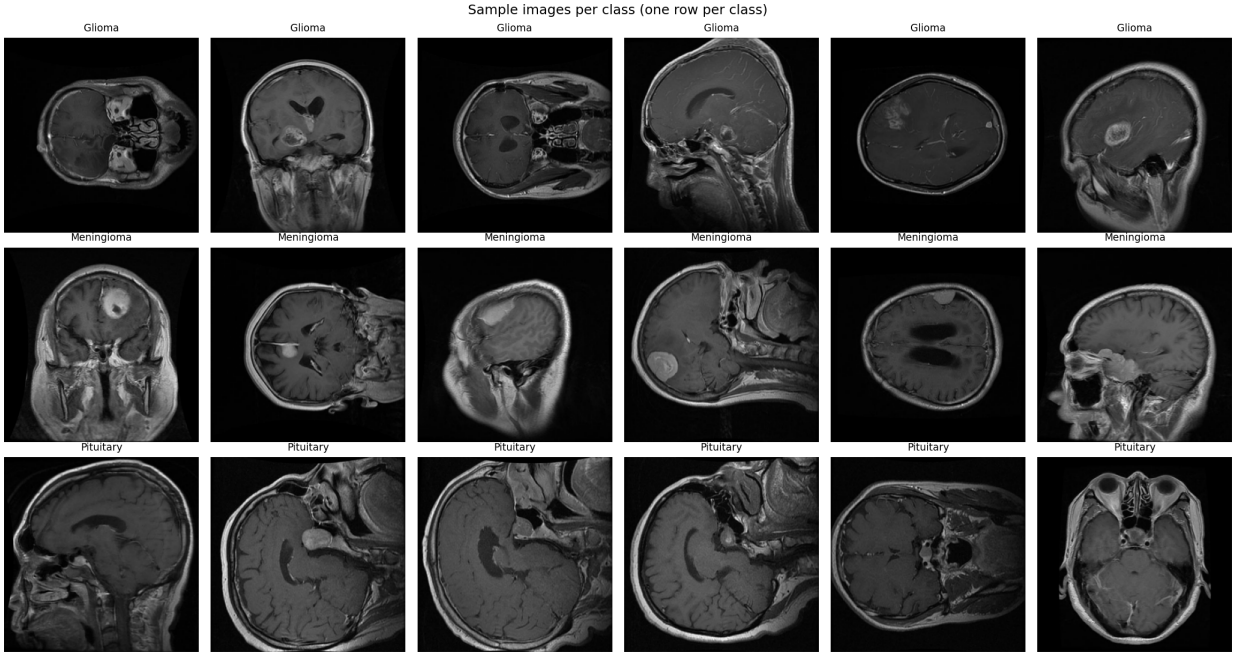


Figure 4: Sample MRI images by tumour type for the multiclass classification task, showing glioma, meningioma, and pituitary cases.

3.3. Preprocessing and Deduplication

When working with multiple datasets there is always a possibility of encountering duplicates, or images of the same underlying content that differ only in resizing or compression. These near-duplicates, if inadvertently distributed across training and test splits, constitute a form of data leakage that artificially inflates reported performance. To address this, we employed perceptual hashing (pHash), which generates a compact fingerprint for each image that is robust to resizing, compression, slight rotation, and minor contrast adjustment. Specifically, 64-bit perceptual hashes were generated using the ImageHash library. The Hamming distance between two hash values served as the similarity metric. We adopted a tolerance threshold of 5, so that any pair of images differing in at most 5 of the 64 bits was treated as a near-duplicate pair, and all but one representative of each such group were removed. The rationale for rigorous deduplication is that multiple public datasets often contain the same underlying images because they derive from common original sources, that near-duplicates cause leakage if present across train and test splits, and that removal of duplicates ensures reported metrics reflect genuine generalisation.

Deduplication was applied to the multiclass collection, where the two constituent datasets are known to overlap substantially. After applying perceptual hashing, the glioma class decreased from 2,352 to 2,000 images, a reduction of 14.97%; meningioma decreased from 1,645 to 1,304 images, a reduction of 20.73%; and pituitary decreased from 1,831 to 1,401 images, a reduction of 23.48%. Overall, the multiclass dataset was reduced from 5,828 to 4,705 images, a reduction of 19.3%. The class distribution before and after deduplication is shown in Figure 5.

It should be stated explicitly that perceptual hash deduplication was *not* applied to the binary detection dataset. The two binary source collections are disjoint in origin and the smaller of the two contributes only 253 of the 3,253 images, so the risk of cross-dataset duplication was judged to be low. This is nevertheless an asymmetry between the two experimental tracks: the multiclass results are protected against near-duplicate leakage by construction, whereas the binary results are not, and the binary figures reported in Section 4.1 should be read with that caveat in mind. Applying the same deduplication procedure to the binary track is identified in Section 5 as a priority for future work.

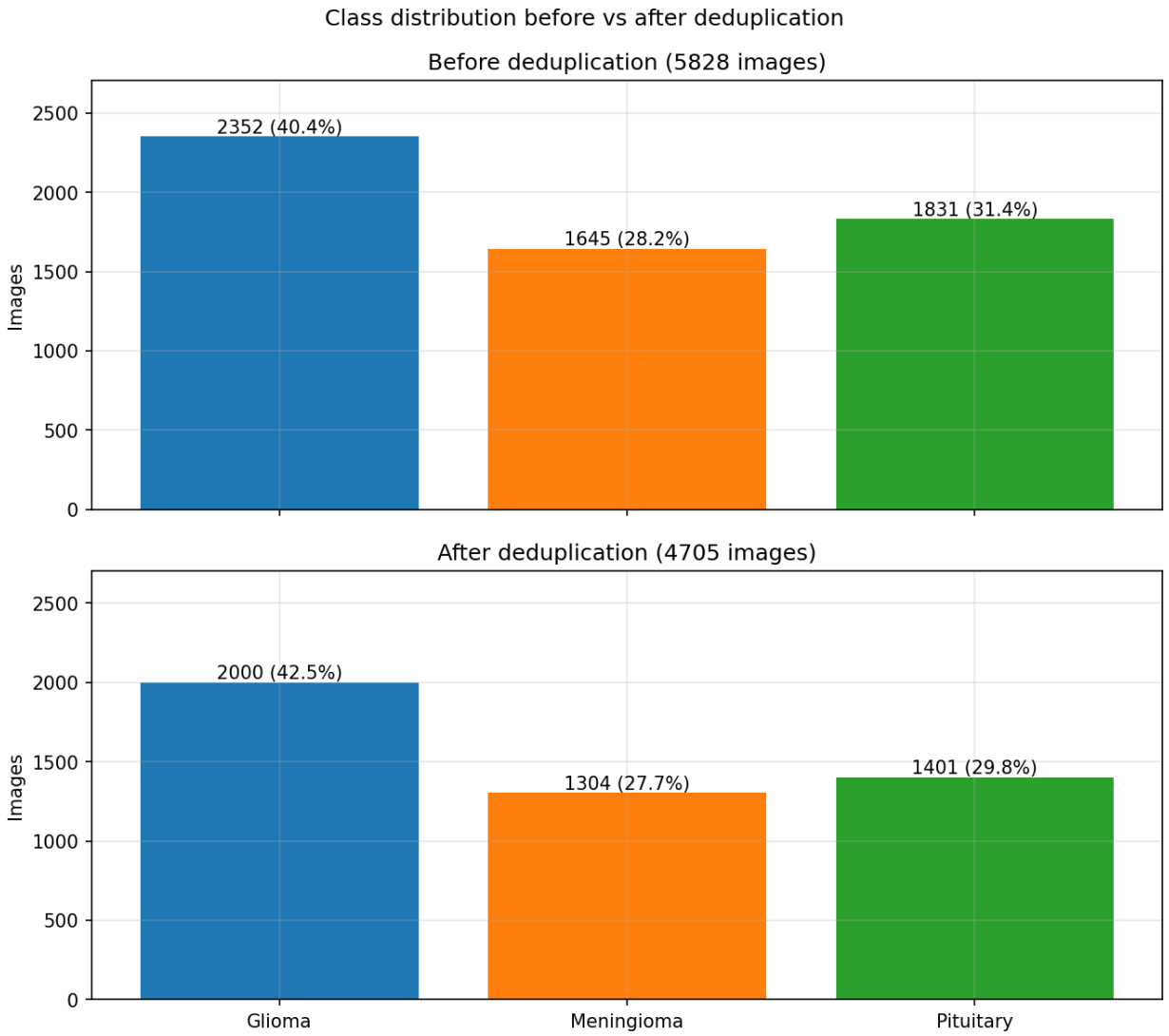


Figure 5: Class distribution of the multiclass dataset before and after perceptual hash deduplication. The total decreased from 5,828 to 4,705 images, a reduction of 19.3%.

Stratified sampling was used for both splits so that every subset retained a representative distribution of the classes. For binary detection, the 3,253 images were divided into training, validation, and test subsets in the proportions 70%, 15%, and 15%, giving 2,277, 488, and 488 images respectively. For multiclass classification, the same proportions were applied to the 4,705 deduplicated images, giving 3,293 training, 706 validation, and 706 test images. The resulting per-class split sizes are reported in Table 3.

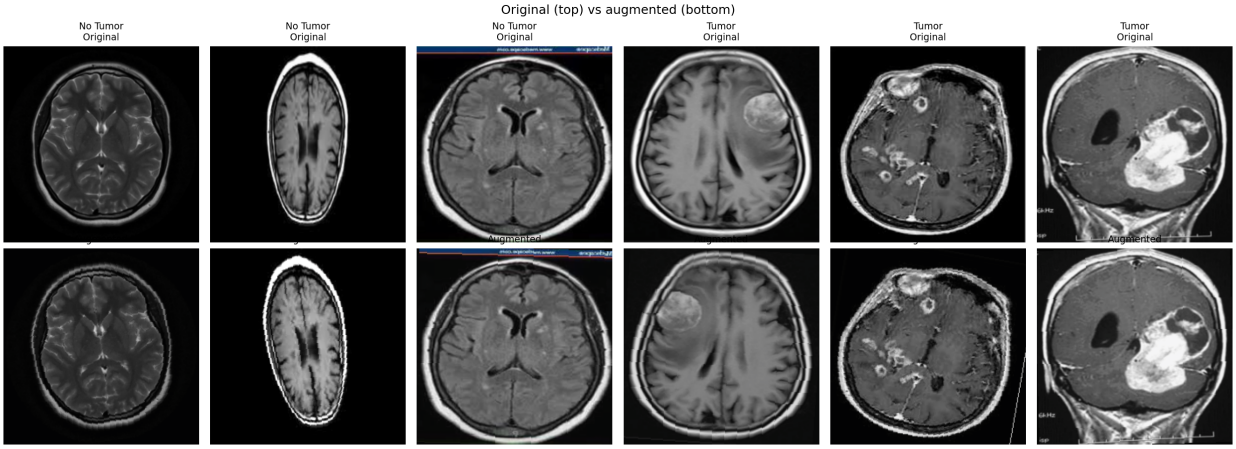
Table 3: Stratified dataset splits for both tasks. Proportions are 70% training, 15% validation, and 15% test in each case.

Task	Class	Training	Validation	Test
Binary detection	No tumour	1,119	239	240
	Tumour	1,158	249	248
	Total	**2,277**	**488**	**488**
Multiclass	Glioma	1,400	300	300
	Meningioma	913	195	196
	Pituitary	980	211	210
	Total	**3,293**	**706**	**706**

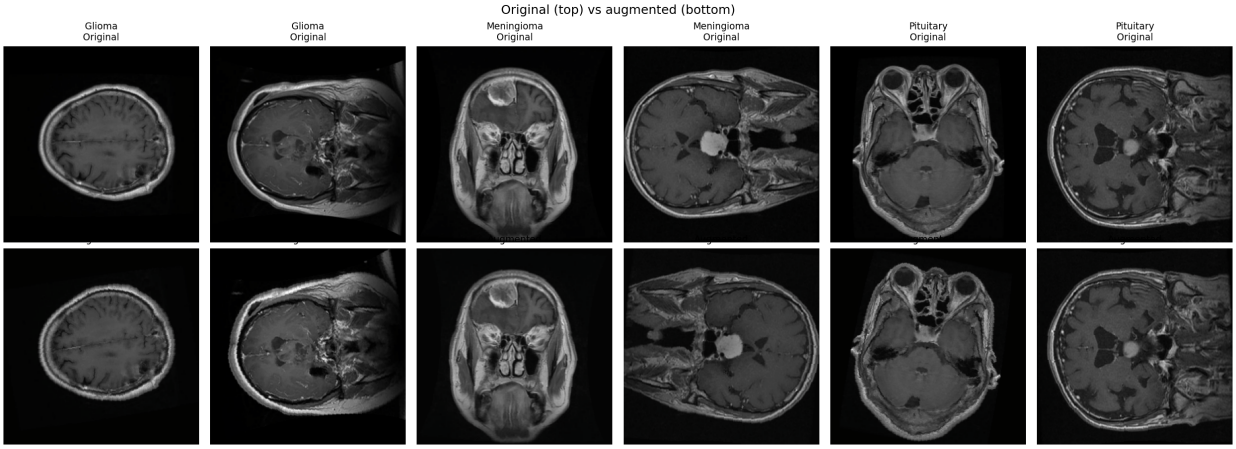
All images were loaded using the Python Imaging Library and converted to the RGB colour format even where the original was greyscale, ensuring compatibility with the ImageNet-pretrained backbones. Images were resized to the input resolution expected by each backbone, which was 224×224 pixels for every model except InceptionV3, which requires 299×299 pixels. Pixel values were converted to a PyTorch tensor and scaled to the range $[0, 1]$, then normalised using the ImageNet channel means of $[0.485, 0.456, 0.406]$ and standard deviations of $[0.229, 0.224, 0.225]$. To improve generalisation, data augmentation was applied exclusively to the training set and consisted of random horizontal flipping with probability 0.5, random rotation of up to ± 10 degrees, and colour jitter adjusting brightness and contrast by up to 20%. These augmentations were applied on the fly at each epoch. For validation and testing, only resizing, tensor conversion, and normalisation were applied, with no stochastic augmentation, ensuring deterministic and reproducible evaluation. Figure 6 illustrates representative images before and after augmentation for both tasks.

3.4. Model Development

Nine architectures spanning three families were evaluated. All were initialised from ImageNet-pretrained weights, and in every case the final classification layer was replaced with a linear layer producing one output for the binary task or three outputs for the multiclass task. The parameter counts quoted below and used throughout the paper are the counts measured directly from the instantiated models after head replacement, and therefore differ slightly from the nominal counts commonly quoted for the ImageNet-1000 variants of the same backbones. These measured values are reported consistently in Tables 6, 8, and 10.



(a) Binary detection task.



(b) Multiclass classification task.

Figure 6: Representative MRI images before and after the training-time augmentation pipeline, comprising random horizontal flipping, random rotation of up to ±10 degrees, and brightness and contrast jitter of up to 20%.

ResNet50 is a 50-layer residual network built around bottleneck residual blocks with skip connections that address the vanishing gradient problem, measured at 23.51 million parameters. VGG16 is a 16-layer network using only 3×3 convolutional filters stacked in increasing depth, measured at 134.26 million parameters. DenseNet121 is a 121-layer densely connected network that concatenates the feature maps of all preceding layers, maximising information flow at only 6.95 million parameters. EfficientNetB0, developed through neural architecture search and compound scaling, achieves strong accuracy with only 4.01 million parameters and 0.77 GFLOPs per image. MobileNetV3-Large uses depth-wise separable convolutions for mobile and embedded applications, measured at 4.20 million parameters and the lowest computational cost in the study at 0.43 GFLOPs per image. InceptionV3 is a 48-layer network with parallel convolutional paths of different kernel sizes within inception modules, capturing multi-scale features at 24.35 million parameters and operating at the higher input resolution of 299×299 pixels. ViT-B/16 divides the input image into 16×16 patches and processes them as a sequence with a transformer encoder, measured at 85.80 million parameters and lacking the convolutional inductive bias of the CNN backbones. Swin-Base introduces hierarchical transformers with shifted windows, achieving linear complexity and multi-scale processing at 86.74 million parameters. CoAtNet-0 combines convolutional and attention mechanisms in a unified architecture and is measured at 26.67 million parameters. We note that CoAtNet-0 is frequently described in the literature as a model of approximately 10 million parameters; the implementation used here is the `coatnet_0_rw_224` variant, whose measured size is 26.67 million parameters, and that measured value is what is reported throughout.

The multiclass track was run for six of the nine architectures, namely CoAtNet-0, EfficientNetB0, InceptionV3, ResNet50, Swin-Base, and ViT-B/16. VGG16, DenseNet121, and MobileNetV3-Large were evaluated on the binary detection task only.

3.5. Training Configuration

All experiments were conducted on dual NVIDIA Tesla T4 graphics processing units with 16 GB of GDDR6 memory each, Intel Xeon processors with four virtual central processing unit cores, and 29 GB of system memory, running Python 3.12.13 with PyTorch 2.10.0 and CUDA 12.8. To ensure fair comparison, the hyperparameters listed in Table 4 were standardised across all models. Every model was trained for 10 epochs using the AdamW optimiser, which is the Adam optimiser with decoupled weight decay regularisation, at an initial learning rate of 3×10^{-4} and the library default weight decay coefficient of 0.01, which was left unchanged rather than tuned. The learning rate was annealed using the cosine annealing learning rate scheduler with the maximum number of iterations parameter set to 10, matching the number of epochs, so that the learning rate followed a single cosine decay from its initial value to approximately zero over the course of training. The loss function was binary cross-entropy with logits loss for the binary detection task, which combines a sigmoid activation with the binary cross-entropy criterion in a numerically stable form, and cross-entropy loss for the multiclass task. Mixed precision training was used throughout, with automatic mixed precision autocasting and gradient scaling to preserve numerical stability in half precision.

Batch sizes were model-specific and set to the largest value that fitted within the available graphics memory, since a single batch size would have been infeasible for the largest transformer backbones. The batch sizes used are listed per model in Tables 10 and 11. This has an important consequence that must be stated plainly: because the learning rate and epoch count are held fixed while the batch size varies from 32 to 256, the number of optimiser steps each model receives varies correspondingly, from roughly 80 to 350 over the ten epochs. The comparison is therefore controlled for data, schedule, and preprocessing, but it is *not* controlled for optimisation budget, and a model trained at batch size 256 receives far fewer gradient updates than one trained at batch size 64. Differences between models consequently confound architecture with optimisation budget to an extent that cannot be separated from these runs alone.

The training loader used `drop_last=True`, so any final partial batch was discarded at each epoch. At batch size 256 this means 2,048 of the 2,277 binary training images were used per epoch, with 229 images dropped; because the loader reshuffles, the dropped subset differs from epoch to epoch, so all images are seen over the course of training but no single epoch sees the full set. The training loop at each epoch consisted of a training phase with augmentation and weight updates followed by a validation phase without augmentation. Learning rate scheduling was applied after each epoch. Model checkpointing retained the weights giving the best validation AUC for the binary task and the best validation macro AUC for the multiclass task, with the first epoch attaining the maximum retained.

The random seed was fixed at 42 for PyTorch, NumPy, and the Python standard library random module. Reproducibility is nevertheless only approximate rather than exact, because `cudnn.benchmark` was enabled, TF32 matrix multiplication was permitted, and automatic mixed precision was used throughout; each of these admits nondeterministic kernel selection. Re-running a notebook should reproduce the reported figures closely but not necessarily bit-for-bit.

Table 4: Training hyperparameters, standardised across all models and both tasks.

Hyperparameter	Setting
Epochs	10
Optimiser	AdamW (Adam with decoupled weight decay)
Weight decay	0.01
Initial learning rate	3×10^{-4}
Learning rate scheduler	Cosine annealing, maximum iterations = 10
Loss (binary detection)	Binary cross-entropy with logits loss
Loss (multiclass)	Cross-entropy loss
Precision	Mixed precision with automatic casting and gradient scaling
Batch size	Model-specific; largest fitting in memory (see Table 10)
Input resolution	224×224 pixels; 299×299 pixels for InceptionV3
Checkpoint selection	Best validation AUC (binary), best validation macro one-versus-rest AUC (multiclass); first epoch attaining the maximum
Random seed	42

3.6. Implementation Differences Between Models

Each architecture was implemented in its own notebook from a shared template, and the intended difference between notebooks is the backbone alone. In practice a small number of further differences were necessary or arose incidentally, and since several of them bear on the interpretation of the results they are documented here rather than left implicit. They are summarised in Table 5.

Two differences are unavoidable. First, each architecture exposes its classification head under a different attribute, so the head replacement code necessarily differs. Second, the layer targeted for gradient-weighted class activation mapping must be the last spatial or token stage of each architecture, which differs by design; for VGG16 this resolves to a max-pooling rather than a convolutional layer, which makes its attention maps less directly comparable with the others.

One difference is an inconsistency that should be taken into account when reading the two tasks together. Most backbones were obtained from `torchvision`; CoAtNet-0 was obtained from `timm` in both tasks, since CoAtNet has no `torchvision` implementation. Swin-Base, however, was not sourced consistently: the binary detection notebook uses the `timm` model `swin_base_patch4_window7_224`, whereas the multiclass notebook uses the `torchvision` model `swin_b`. These are different implementations of the Swin-B architecture and they load different pretrained checkpoints, so the two rows labelled Swin-Base in Tables 6 and 8 are not the same model initialised the same way. Their measured parameter counts differ only marginally, at 86.74 and 86.75 million, but the comparison of Swin-Base across the two tasks is not strictly like-for-like, and the contrast between its weak binary result of 95.90% and its leading multiclass result of 98.58% should not be attributed to task difficulty alone.

Two differences affect results and deserve particular attention. The InceptionV3 binary notebook applies the standard auxiliary-classifier recipe, adding 0.4 times the auxiliary loss to the main loss during training, which no other model receives. Separately, although the InceptionV3 binary model is wrapped for two-GPU data parallelism, the training loop calls the unwrapped module, so the run executed on a single GPU; the recorded peak memory of 0 MB on the second device confirms this. Its training time in Table 10 is therefore a single-GPU measurement and is not comparable with the other rows, a point returned to in Section 4.3. In the multiclass InceptionV3 notebook the auxiliary head is disabled but not removed, so 769,000 parameters belonging to an unused 1000-class auxiliary classifier remain in the model and are counted in its reported total of 25.12 million. This accounts for the difference between the binary and multiclass InceptionV3 parameter counts in Tables 6 and 8, and it is the reason those two figures are not identical despite the backbone being the same.

Table 5: Implementation differences between the per-model notebooks, with the reason for each. Differences confined to comments, cell titles, or equivalent reformatting are not listed.

Difference	Scope	Reason
Head replacement code	All models	Each architecture exposes its classifier under a different attribute; unavoidable.
Source library `timm`	CoAtNet-0 (both tasks)	CoAtNet has no `torchvision` implementation.
Source library differs *between tasks*	Swin-Base	`timm` `swin_base_patch4_window7_224` for binary, `torchvision` `swin_b` for multiclass. Different implementations and different pretrained checkpoints; no stated reason.
Input resolution 299×299	InceptionV3	The architecture's native input size.
Batch size (32–256)	All models	Set to the largest value fitting in 16 GB of graphics memory. Only InceptionV3 states this in code; for the others it is inferred from recorded peak memory. Changes the optimisation budget.
Auxiliary loss 0.4×	InceptionV3 (binary)	Standard Inception training recipe; applied without comment.
Single-GPU execution	InceptionV3 (binary)	The training loop calls the unwrapped module, bypassing data parallelism. Unintended.
Retained unused auxiliary head	InceptionV3 (multiclass)	Only the main head was replaced; adds 769k parameters to the reported total. Unintended.
Grad-CAM target layer	All models	Must be the final spatial or token stage of each architecture. For VGG16 this is a max-pooling layer.
Pinned `IMAGENET1K_V1` weights	MobileNetV3-Large	The only model not using the library default weights. No stated reason.

A further asymmetry concerns the deduplication procedure itself. The perceptual hash pass operates over the combined multiclass pool without regard to class label, and processes classes in a fixed order, so where a near-duplicate pair spans two classes the glioma image is always the one retained. Such cross-label removals are neither counted nor reported by the procedure. The class-wise reduction percentages in Section 3.3 should therefore be read as the outcome of an order-dependent process rather than as an intrinsic measure of redundancy within each class.

Finally, the multiclass source collection was assembled by merging the official training and testing folders of the Bhuvaji dataset into a single pool before applying deduplication and drawing new stratified splits. This was done so that both tasks would use the same 70/15/15 splitting procedure, but it means the multiclass results are not directly comparable with published figures that evaluate on that dataset's original test split, a point returned to in Section 4.5.

3.7. Evaluation Metrics

All results reported in Section 4 were computed on the held-out test split, which was evaluated exactly once per model using the checkpoint selected on the validation split. Validation-split results are reported separately in Appendix A and Appendix B for transparency.

Overall performance is summarised by accuracy and, for the multiclass task, by balanced accuracy, which averages the per-class recall and is therefore not dominated by the larger glioma class. Per-class behaviour is reported through precision, recall or sensitivity, specificity, and the F1 score, since a headline accuracy figure alone conceals the asymmetry between missing a tumour and raising a false alarm. For the binary task, sensitivity is the proportion of tumour-positive images correctly identified and specificity is the proportion of tumour-negative images correctly identified; in a screening context a false negative is the more costly error of the two. Ranking quality independent of any decision threshold is captured by the area under the receiver operating characteristic curve, which ranges from 0.5 for random ranking to 1.0 for perfect separation; the one-versus-rest formulation averaged over classes is used for the multiclass task. Probability quality is assessed by the cross-entropy test loss and by the expected calibration error, which measures the average absolute gap between predicted confidence and observed accuracy across confidence bins, and which matters when a predicted probability is to be presented to a clinician as a degree of confidence. The test loss quoted in Tables 6 and 8 is the log loss evaluated in double precision over a single-device re-inference pass across the test split, rather than the running mean of the per-batch losses printed during training. Both passes compute the forward step under half-precision autocasting, but the re-inference pass accumulates the resulting probabilities in double precision and renormalises them before the metric is computed, whereas the training-time figure averages half-precision batch losses. The two agree to within roughly 10^{-3} and, in fourteen of the fifteen runs, agree exactly on accuracy; the re-inference figure is the more precise quantity and is the one reported throughout. Agreement corrected for chance is reported through the Matthews correlation coefficient and Cohen's kappa. Efficiency is characterised by measured parameter count, floating point operations per image, wall-clock training time, peak graphics memory, single-image inference latency, and batched throughput, all measured on the same hardware.

4. Results and Discussion

This section presents the findings of the binary detection and multiclass classification experiments. Section 4.1 reports binary tumour detection, Section 4.2 reports classification into glioma, meningioma, and pituitary subtypes, Section 4.3 compares computational efficiency, Section 4.4 examines model interpretability through gradient-weighted class activation mapping, and Section 4.5 compares the present results with previously published work. Unless stated otherwise, every figure quoted is measured on the held-out test split.

4.1. Binary Detection Results

Test-split results for the binary detection task are reported in Table 6. Four architectures tied at the highest accuracy of 99.39%, corresponding to three misclassified images out of 488: CoAtNet-0, DenseNet121, EfficientNetB0, and ResNet50. Although the accuracy of these four models is identical, the structure of their errors differs in a way that matters clinically. CoAtNet-0 and EfficientNetB0 achieved perfect sensitivity of 100.00% at a specificity of 98.75%, meaning that every tumour-positive image in the test set was detected and the three errors were false alarms on tumour-negative images. DenseNet121 and ResNet50 showed the mirror-image behaviour, achieving perfect specificity of 100.00% at a sensitivity of 98.79%, so their three errors were missed tumours. For a screening application in which a missed tumour is the more serious failure, the error profile of CoAtNet-0 and EfficientNetB0 is preferable to that of DenseNet121 and ResNet50 despite the identical headline accuracy. DenseNet121 and ResNet50 nevertheless produced the strongest threshold-independent ranking, both reaching an AUC of 1.0000 to four decimal places, indicating that a different operating threshold would have separated the two classes almost perfectly.

These four models should be regarded as statistically indistinguishable rather than genuinely tied. On a test set of 488 images a single misclassification is worth 0.20 percentage points of accuracy, each model was trained once from a single seed, and no confidence intervals are available; the separation between the four leaders and InceptionV3 amounts to three images. The ranking at the top of Table 6 should therefore be read as identifying a group of models that perform equivalently on this data, not as ordering them. The distinction between the group and the weaker models, by contrast, spans tens of images and is not in doubt.

InceptionV3 followed at 98.77% accuracy with an AUC of 0.9986. The three transformer and hybrid models diverged sharply: CoAtNet-0, which combines convolution with attention, matched the best CNNs, whereas the two pure transformers trailed, with Swin-Base at 95.90% and ViT-B/16 at 94.88%. MobileNetV3-Large reached only 90.78% accuracy despite an AUC of 0.9981, and this gap between ranking quality and accuracy is diagnostic: the model ranks images well but its decision threshold is badly placed, as confirmed by its sensitivity of 81.85% against a specificity of 100.00% and by its expected calibration error of 0.0685, the worst in the study. The extent of the collapse is shown by its optimal Youden threshold of 0.00044, some three orders of magnitude below the 0.5 decision point actually used, so that 45 tumours are missed purely through threshold placement. Recalibration alone would recover most of the loss, and the model's poor accuracy should not be read as an inability to discriminate. Its validation loss of 0.3995 against a training accuracy of 99.90% at the final epoch indicates substantial overfitting. VGG16 performed worst at 88.32% accuracy with an AUC of 0.9676, consistent with the expectation that a 134-million-parameter architecture without residual or dense connectivity is poorly suited to a dataset of this size.

Table 6: Binary detection results on the held-out test split (488 images). Sensitivity is recall of the tumour-positive class and specificity is recall of the tumour-negative class. Parameter counts are measured from the instantiated models. Rows are ordered by accuracy.

Model	Acc. (%)	Sens. (%)	Spec. (%)	F1 (%)	AUC	Test loss	Params (M)
CoAtNet-0	99.39	100.00	98.75	99.40	0.9999	0.0253	26.67
DenseNet121	99.39	98.79	100.00	99.39	1.0000	0.0189	6.95
EfficientNetB0	99.39	100.00	98.75	99.40	0.9998	0.0233	4.01
ResNet50	99.39	98.79	100.00	99.39	1.0000	0.0217	23.51
InceptionV3	98.77	97.98	99.58	98.78	0.9986	0.0419	24.35
Swin-Base	95.90	93.55	98.33	95.87	0.9961	0.1117	86.74
ViT-B/16	94.88	96.37	93.33	95.03	0.9862	0.1525	85.80
MobileNetV3-Large	90.78	81.85	100.00	90.02	0.9981	0.3698	4.20
VGG16	88.32	86.29	90.42	88.25	0.9676	0.2448	134.26

The per-class breakdown for every model is given in Table 7, which makes the error asymmetry described above explicit. EfficientNetB0 is used for the detailed per-model figures in this subsection, because it offers the best accuracy-to-cost ratio and is the checkpoint released with the accompanying code. Its training and validation curves are shown in Figure 7, its confusion matrix in Figure 8, its receiver operating characteristic and precision–recall curves in Figure 10, its per-class metric summary in Figure 11, its reliability diagram in Figure 12, and representative test-set predictions with their assigned probabilities in Figure 13.

The error asymmetry is seen most directly by comparing Figure 8 with Figure 9, which shows the confusion matrices of ResNet50 at the same accuracy of 99.39%. The two matrices are mirror images of one another: the three EfficientNetB0 errors all fall in the false-positive cell, so no tumour-positive image is missed, whereas the three ResNet50 errors all fall in the false-negative cell, so three tumours are missed and no healthy scan is flagged. This is the distinction that a single accuracy figure hides, and it is the one that would determine which model is appropriate for a screening deployment. Equivalent figures for all remaining models are generated by the released notebooks and are available in the code repository described in Appendix C.

Table 7: Per-class binary detection metrics on the held-out test split. Support is the number of test images in each class.

Model	Class	Support	Prec. (%)	Sens. (%)	Spec. (%)	F1 (%)
CoAtNet-0	No tumour	240	100.00	98.75	100.00	99.37
	Tumour	248	98.80	100.00	98.75	99.40
DenseNet121	No tumour	240	98.77	100.00	98.79	99.38
	Tumour	248	100.00	98.79	100.00	99.39
EfficientNetB0	No tumour	240	100.00	98.75	100.00	99.37
	Tumour	248	98.80	100.00	98.75	99.40
ResNet50	No tumour	240	98.77	100.00	98.79	99.38
	Tumour	248	100.00	98.79	100.00	99.39
InceptionV3	No tumour	240	97.95	99.58	97.98	98.76
	Tumour	248	99.59	97.98	99.58	98.78
Swin-Base	No tumour	240	93.65	98.33	93.55	95.93
	Tumour	248	98.31	93.55	98.33	95.87
ViT-B/16	No tumour	240	96.14	93.33	96.37	94.71
	Tumour	248	93.73	96.37	93.33	95.03
MobileNetV3-Large	No tumour	240	84.21	100.00	81.85	91.43
	Tumour	248	100.00	81.85	100.00	90.02
VGG16	No tumour	240	86.45	90.42	86.29	88.39
	Tumour	248	90.30	86.29	90.42	88.25

4.2. Multiclass Classification Results

Test-split results for classification into glioma, meningioma, and pituitary subtypes are reported in Table 8. Swin-Base achieved the highest accuracy at 98.58%, corresponding to 10 misclassified images out of 706, with a balanced accuracy of 98.42%, a macro F1 score of 98.47%, and the lowest test loss in the study at 0.0410. CoAtNet-0 followed at 98.44% and InceptionV3 at 98.30%, the latter achieving the highest macro one-versus-rest AUC at 0.9994. EfficientNetB0 and ResNet50 tied at 97.59% accuracy, which is a notable result given that EfficientNetB0 uses 4.01 million parameters against 23.51 million for ResNet50, a difference of almost six times for identical accuracy. ViT-B/16 was again the weakest model by a substantial margin at 90.51% accuracy and a macro AUC of 0.9759.

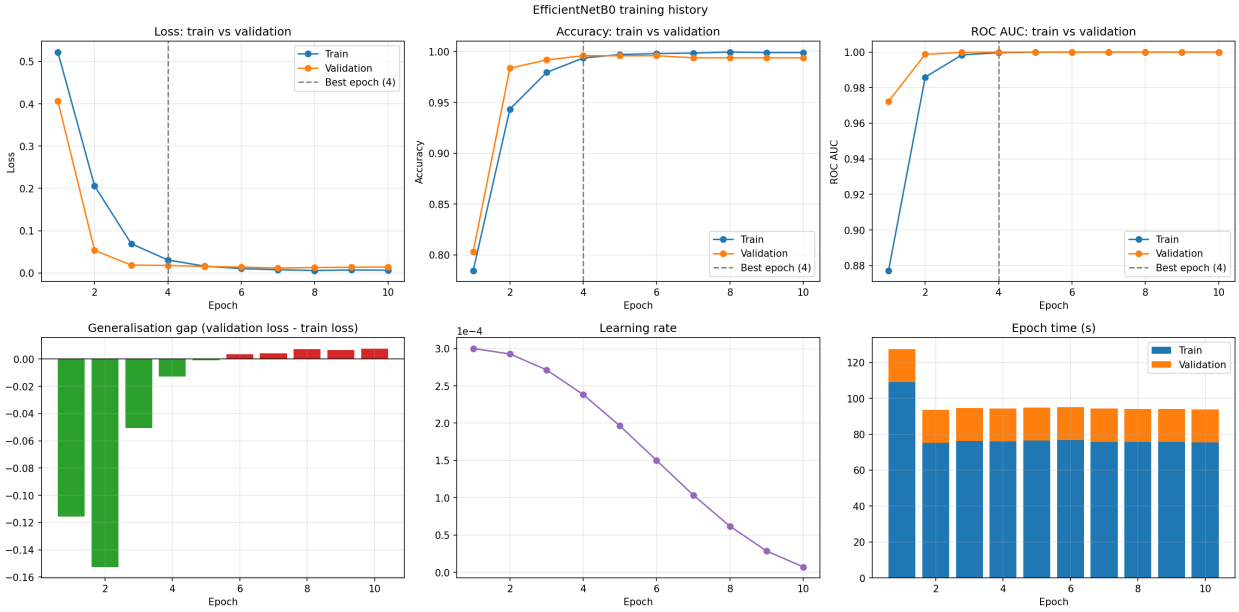


Figure 7: Training and validation curves for **EfficientNetB0** on the binary detection task, showing loss, accuracy, and AUC over 10 epochs. The best validation AUC of 0.9999 was reached at epoch 4.

The contrast between Swin-Base and ViT-B/16 is the clearest architectural signal in these results. Both are transformer backbones of almost identical size, 86.75 million and 85.80 million parameters respectively, yet they differ by just over eight percentage points in accuracy. The difference is that Swin-Base imposes a hierarchical, locally windowed attention structure that reintroduces a spatial inductive bias, whereas ViT-B/16 treats the image as a flat sequence of patches and must learn that structure from data. With only 3,293 training images available, there is insufficient data to learn it, and the pure transformer underperforms. CoAtNet-0 supports the same reading from the other direction: by combining convolutional stages with attention stages it reaches 98.44% with under a third of the parameters of either pure transformer. The conclusion is not that attention is unhelpful for this task but that, at this data scale, attention needs to be paired with a convolutional prior. This reading should be qualified by the interpretability evidence in Section 4.4, which indicates that part of the Swin-Base advantage may rest on non-anatomical image cues; the architectural argument and the shortcut explanation are not mutually exclusive, and these experiments cannot separate them.

The per-class breakdown in Table 9 shows that meningioma is consistently the hardest class for every model. Swin-Base, CoAtNet-0, and EfficientNetB0 all reached a meningioma sensitivity of 95.92%, well below their glioma and pituitary sensitivities, and meningioma carries the lowest F1 score for five of the six models. This is clinically plausible, since meningiomas arise from the meningeal layers and can present with less distinctive intensity and shape signatures on a single two-dimensional slice than the typically well-localised pituitary lesions. Meningioma is also the smallest class, with 913 training images against 1,400 for glioma, so the difficulty reflects both intrinsic similarity and reduced representation. Pituitary tumours were the easiest class overall, with Swin-Base achieving perfect sensitivity of 100.00% on this class.

The confusion matrices of Swin-Base, the best-performing subtype classifier, are shown in Figure 14 and its one-versus-rest receiver operating characteristic and precision–recall curves in Figure 15. The test-split matrix confirms the pattern described above: of the 10 errors, 8 are meningioma cases assigned to another class, while pituitary cases are classified without error.

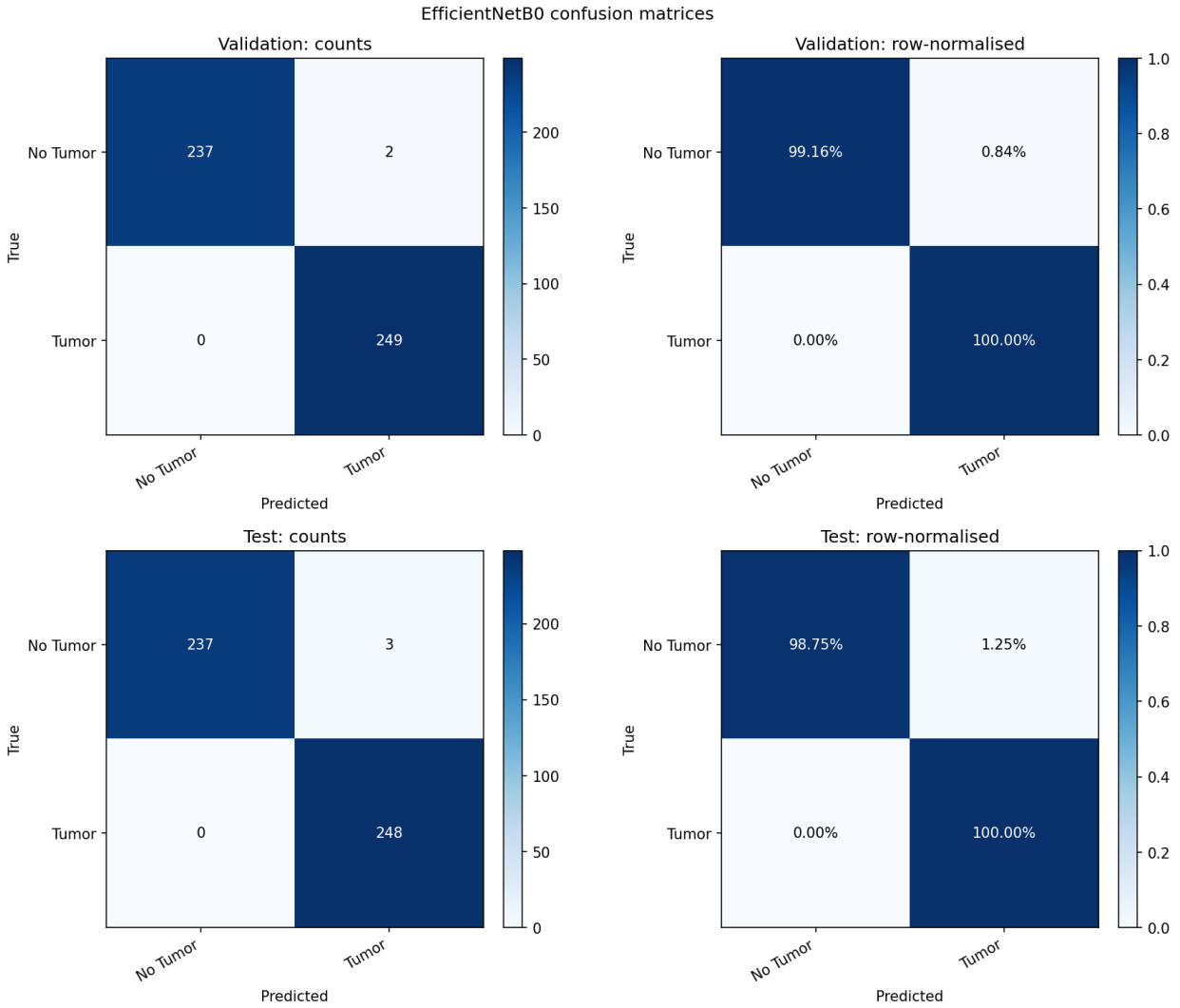


Figure 8: Confusion matrices for **EfficientNetB0** on the binary detection task, shown for the validation and held-out test splits. On the test split the three errors are false positives on tumour-negative images; no tumour-positive image was missed.

Table 8: Multiclass classification results on the held-out test split (706 images). Precision and F1 are macro-averaged across the three tumour classes, and AUC is the macro-averaged one-versus-rest value. Rows are ordered by accuracy.

Model	Acc. (%)	Bal. acc. (%)	Prec. (%)	F1 (%)	AUC	Test loss	Params (M)
Swin-Base	98.58	98.42	98.54	98.47	0.9993	0.0410	86.75
CoAtNet-0	98.44	98.21	98.50	98.35	0.9989	0.0521	26.67
InceptionV3	98.30	98.29	98.17	98.22	0.9994	0.0471	25.12
EfficientNetB0	97.59	97.50	97.41	97.45	0.9988	0.0588	4.01
ResNet50	97.59	97.57	97.36	97.46	0.9988	0.0593	23.51
ViT-B/16	90.51	90.37	90.31	90.34	0.9759	0.2637	85.80

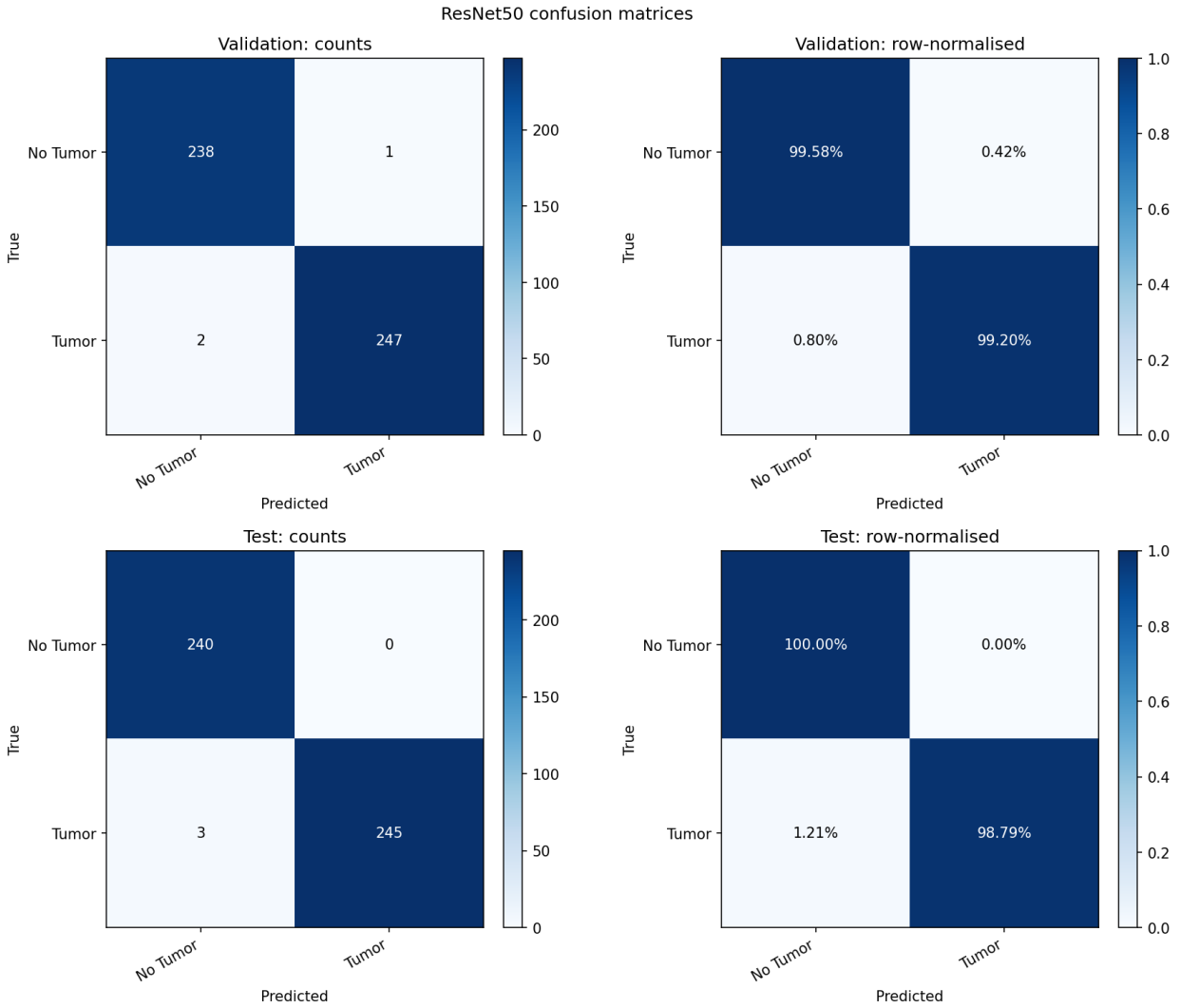


Figure 9: Confusion matrices for **ResNet50** on the binary detection task, shown for the validation and held-out test splits. Compared with EfficientNetB0 in Figure 8, the test-split errors are the mirror image: all three are missed tumours rather than false alarms, at identical overall accuracy.

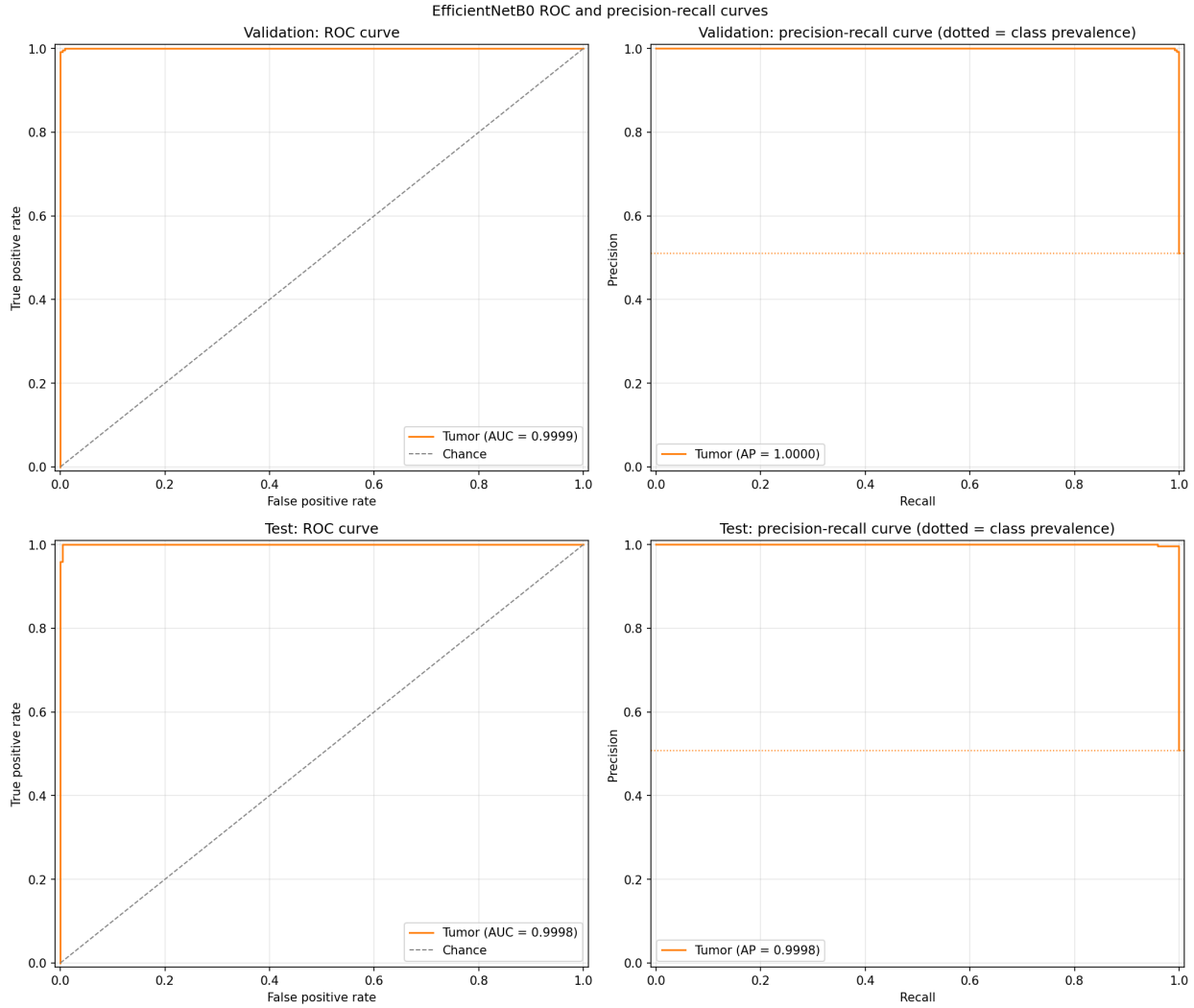


Figure 10: Receiver operating characteristic and precision–recall curves for **EfficientNetB0** on the binary detection test split, corresponding to an AUC of 0.9998 and an average precision of 0.9998.

The corresponding figures for EfficientNetB0 are provided alongside, since it is the released checkpoint and the most practical candidate for constrained deployment. Its training and validation curves are shown in Figure 16, its confusion matrix in Figure 17, its one-versus-rest curves in Figure 18, its reliability diagram in Figure 19, its per-class metric summary in Figure 20, and representative predictions in Figure 21. Comparing Figures 14 and 17 quantifies the cost of that efficiency choice: 10 errors against 17 on the same 706 test images, a difference of seven images for a twenty-onefold reduction in parameter count. The corresponding artefacts for the remaining four multiclass models are produced by the released notebooks.

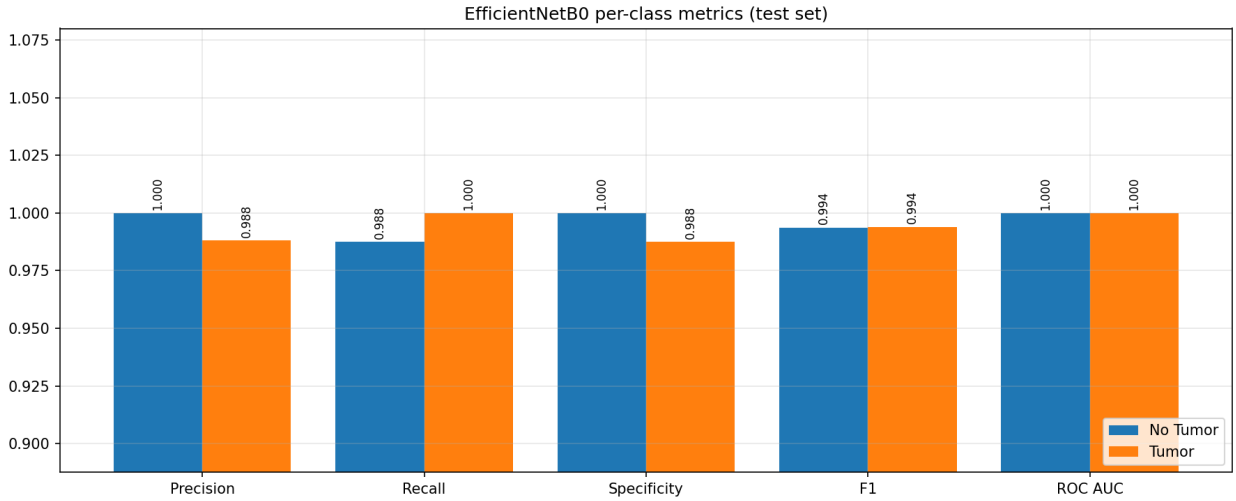


Figure 11: Per-class precision, recall, and F1 scores for **EfficientNetB0** on the binary detection test split. The tumour class attains perfect recall, with the residual error confined to precision on that class.

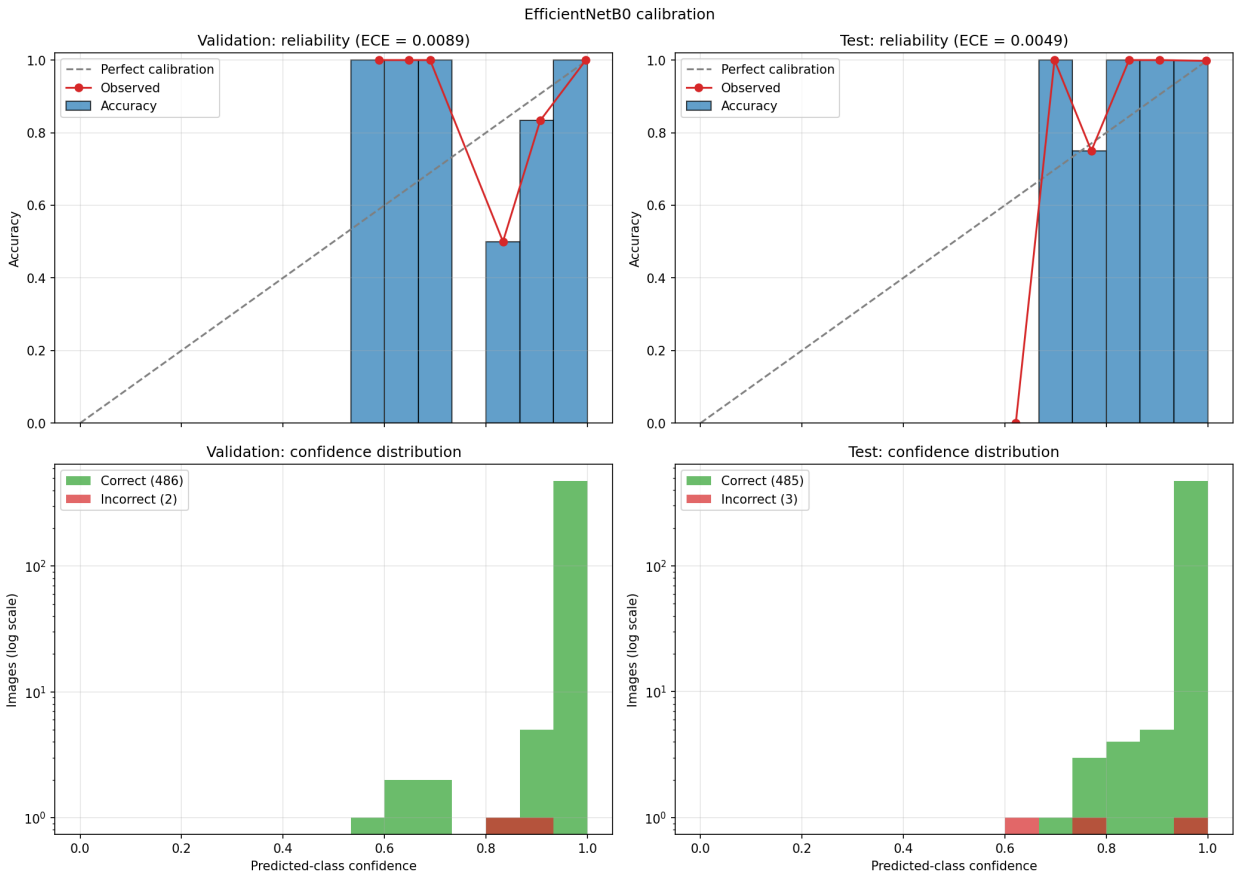


Figure 12: Reliability diagram for **EfficientNetB0** on the binary detection test split, corresponding to an expected calibration error of 0.0049.

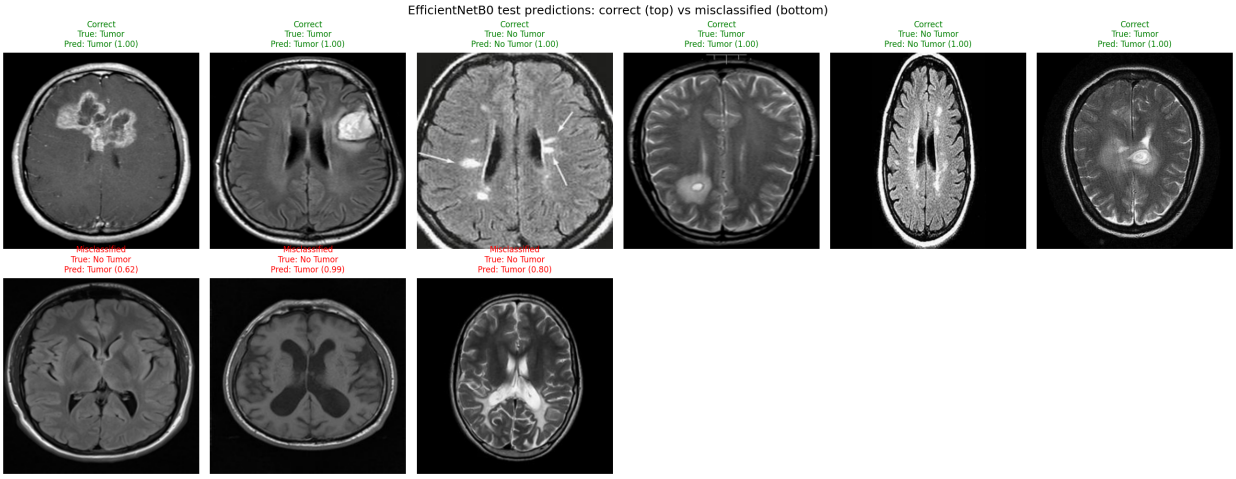


Figure 13: Representative test-set predictions of **EfficientNetB0** on the binary detection task, with the predicted class and associated probability shown for each image.

Table 9: Per-class multiclass classification metrics on the held-out test split. Support is the number of test images in each class.

Model	Class	Support	Prec. (%)	Sens. (%)	Spec. (%)	F1 (%)
Swin-Base	Glioma	300	99.00	99.33	99.26	99.17
	Meningioma	196	98.95	95.92	99.61	97.41
	Pituitary	210	97.67	100.00	98.99	98.82
CoAtNet-0	Glioma	300	98.03	99.67	98.52	98.84
	Meningioma	196	98.43	95.92	99.41	97.16
	Pituitary	210	99.05	99.05	99.60	99.05
InceptionV3	Glioma	300	98.99	98.33	99.26	98.66
	Meningioma	196	96.00	97.96	98.43	96.97
	Pituitary	210	99.52	98.57	99.80	99.04
EfficientNetB0	Glioma	300	98.66	98.00	99.01	98.33
	Meningioma	196	95.92	95.92	98.43	95.92
	Pituitary	210	97.64	98.57	98.99	98.10
ResNet50	Glioma	300	98.99	97.67	99.26	98.32
	Meningioma	196	94.53	96.94	97.84	95.72
	Pituitary	210	98.56	98.10	99.40	98.33
ViT-B/16	Glioma	300	91.28	90.67	93.60	90.97
	Meningioma	196	83.50	85.20	93.53	84.34
	Pituitary	210	96.15	95.24	98.39	95.69

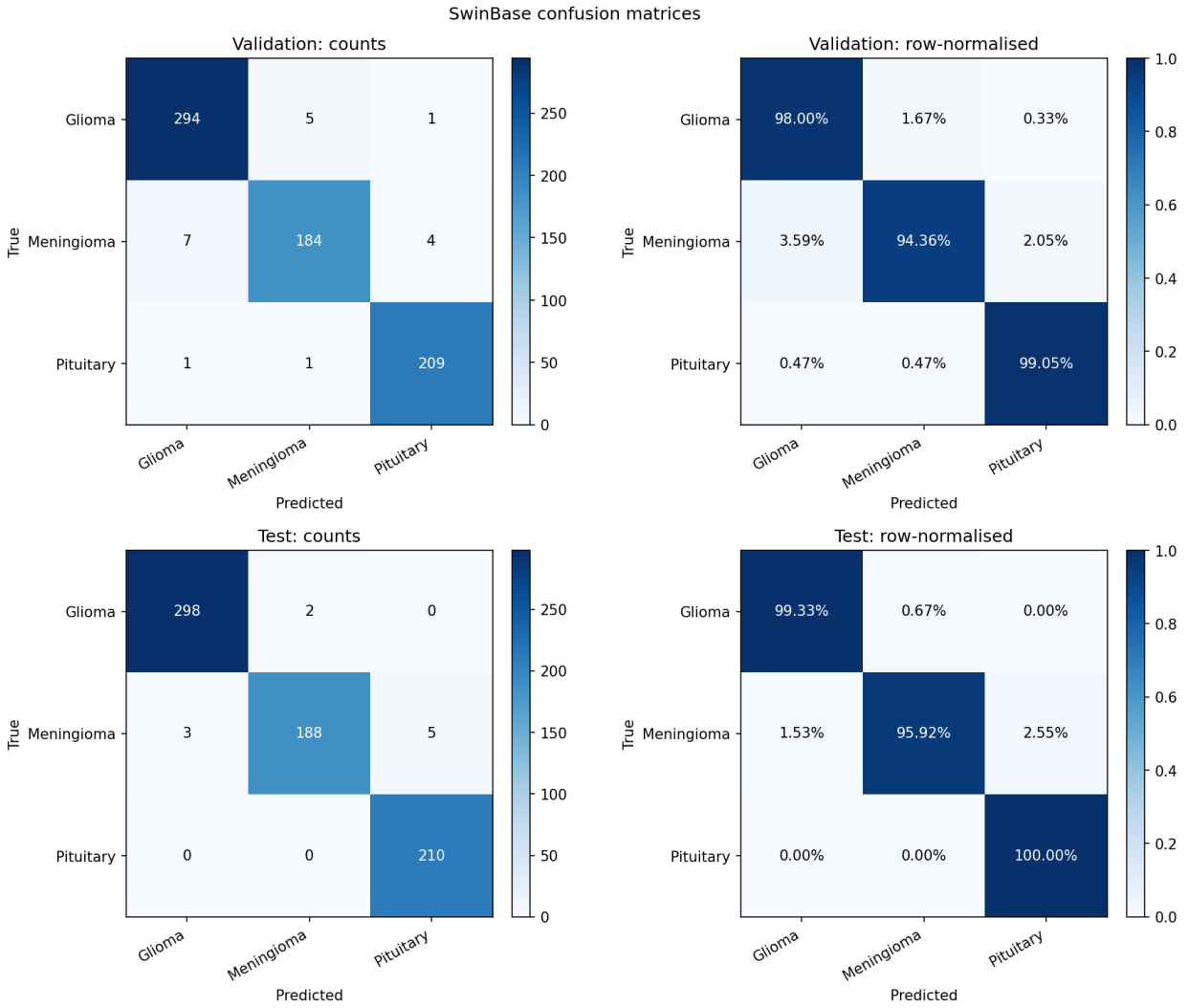


Figure 14: Confusion matrices for **Swin-Base**, the best-performing subtype classifier, on the validation and held-out test splits. Of the 10 test-split errors, 8 involve meningioma; no pituitary case is misclassified.

4.3. Model Efficiency Comparison

Table 10 presents the computational efficiency comparison for the binary detection runs, with all measurements taken on the same dual Tesla T4 configuration. Three distinct efficiency regimes are visible. The lightweight CNNs, EfficientNetB0 and MobileNetV3-Large, require under 0.8 GFLOPs per image and deliver the highest batched throughput, at 1,364 and 2,409 images per second respectively. The mid-sized CNNs, DenseNet121, ResNet50, InceptionV3, and CoAtNet-0, occupy an intermediate band of 5.7 to 11.4 GFLOPs. The large transformers and VGG16 sit at 22.5 to 30.9 GFLOPs and deliver the lowest throughput, with Swin-Base managing only 196 images per second, roughly one twelfth of MobileNetV3-Large.

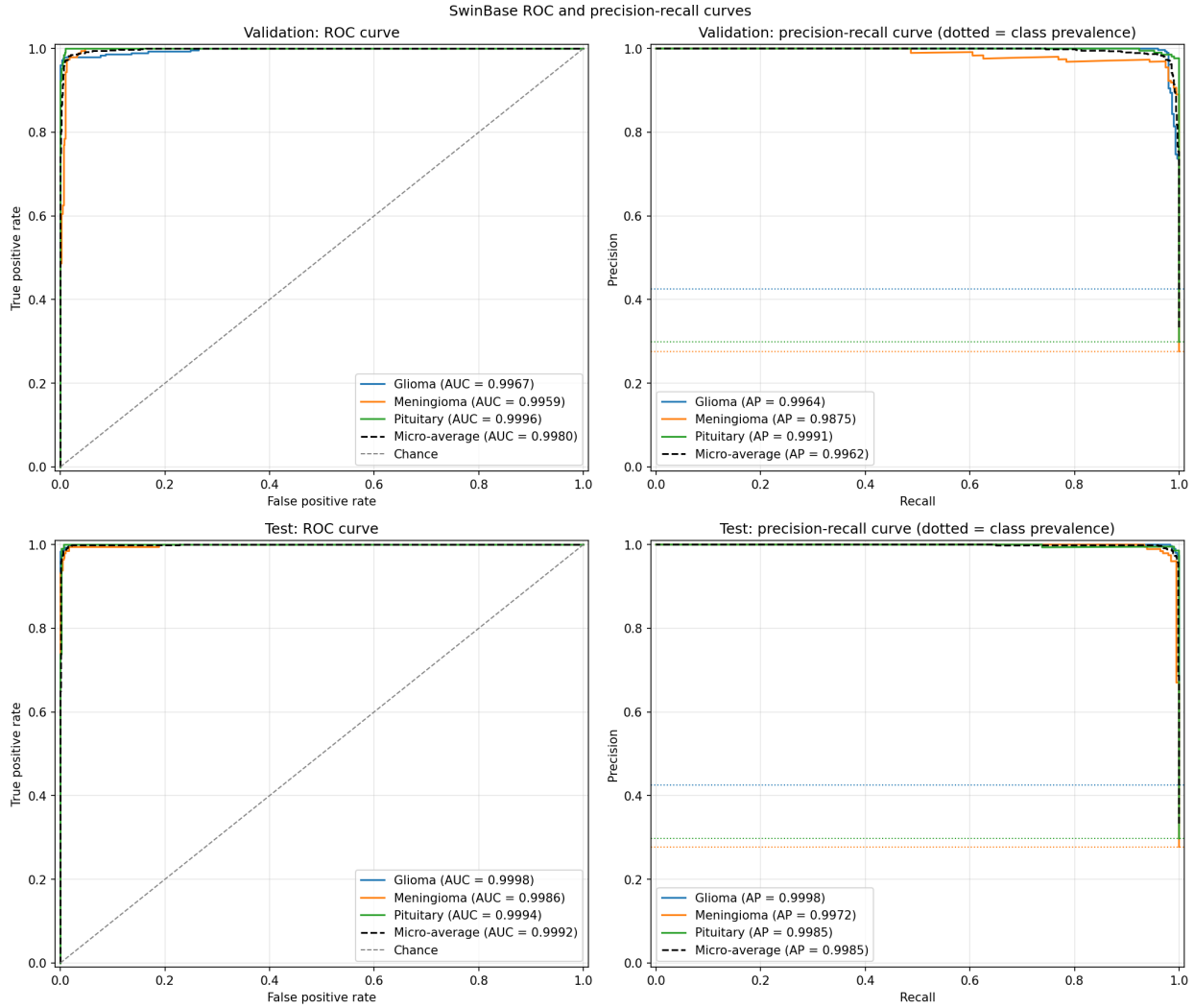


Figure 15: One-versus-rest receiver operating characteristic and precision–recall curves for **Swin-Base** on the multiclass test split, corresponding to a macro-averaged AUC of 0.9993.

The accuracy-per-million-parameters column makes the architectural efficiency difference stark. EfficientNetB0 achieves 24.79 percentage points of accuracy per million parameters, MobileNetV3-Large 21.60, and DenseNet121 14.29, while the transformers deliver 1.11 and VGG16 only 0.66. EfficientNetB0 is therefore more than thirty-seven times more parameter-efficient than VGG16 while also being eleven percentage points more accurate. It is worth noting that wall-clock training time does not track parameter count, because batch size, input resolution, and attention cost all intervene: CoAtNet-0 required 29.1 minutes while EfficientNetB0 required 16.3 minutes despite being the smallest model in the study, a consequence of CoAtNet-0 running at batch size 64 against 256 and therefore taking four times as many optimiser steps. The InceptionV3 training time of 4.9 minutes in Table 10 should be disregarded for comparison purposes: as described in Section 3.6, that run executed on a single GPU rather than two, so its wall-clock figure is not measured under the same conditions as the other eight rows. Its parameter count, GFLOPs, latency, and throughput are unaffected, since those were measured separately on one device for every model. Single-image latency likewise favours ResNet50 at 7.3 ms over EfficientNetB0 at 10.8 ms, because the depth-wise separable convolutions that make EfficientNetB0 parameter-efficient do not map as efficiently onto the GPU at batch size one. For deployment, therefore, the correct efficiency metric depends on whether the constraint is memory footprint, batch throughput, or single-request latency.

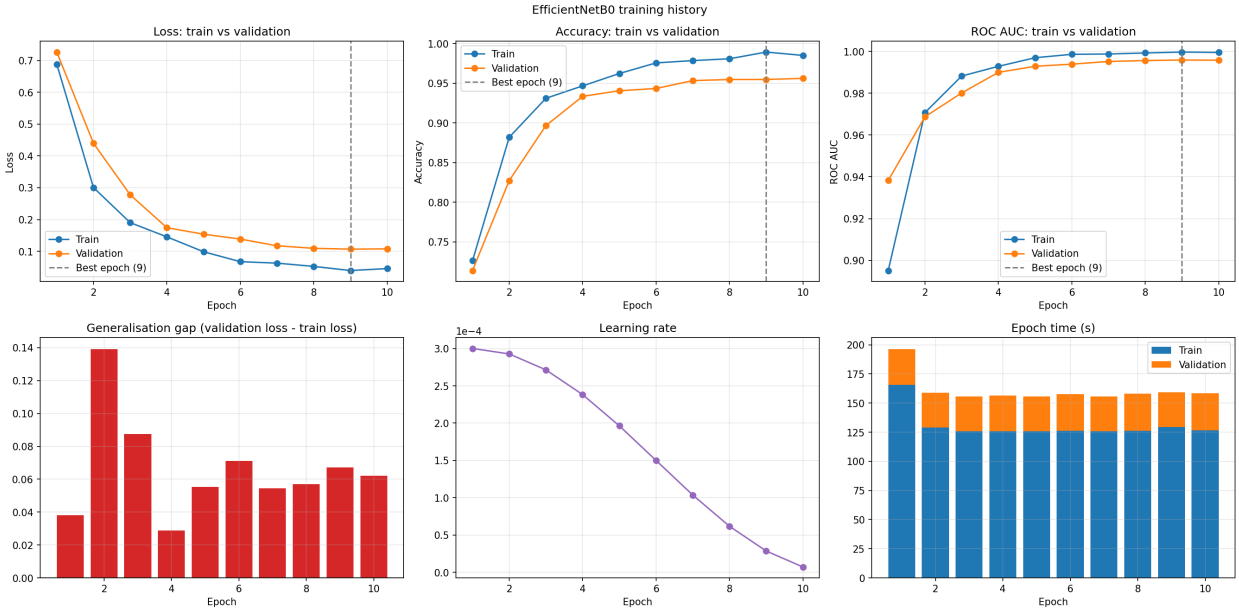


Figure 16: Training and validation curves for **EfficientNetB0** on the multiclass classification task, showing loss, accuracy, and AUC over 10 epochs. The checkpointed epoch is 9, selected on validation macro AUC, at which validation accuracy was 95.47%.

Table 10: Computational efficiency comparison for the binary detection task. Training used dual NVIDIA Tesla T4 graphics processing units, except InceptionV3, which ran on a single device (Section 3.6) and whose training time is therefore not comparable. Latency and throughput were measured on one device for every model, at batch size 1 and 32 respectively. "Thr." is throughput in images per second and "Acc./M" is test accuracy in percentage points per million parameters. Rows are ordered by parameter count.

Model	Params (M)	GFLOPs	Batch	Train (min)	Latency (ms)	Thr. (img/s)	Acc./M
EfficientNetB0	4.01	0.77	256	16.3	10.76	1364	24.79
MobileNetV3-Large	4.20	0.43	256	12.0	8.08	2409	21.60
DenseNet121	6.95	5.67	64	8.5	19.43	630	14.29
ResNet50	23.51	8.17	256	5.4	7.35	771	4.23
InceptionV3	24.35	11.42	128	4.9	14.05	514	4.06
CoAtNet-0	26.67	8.42	64	29.1	13.91	520	3.73
ViT-B/16	85.80	22.54	128	4.7	8.80	287	1.11
Swin-Base	86.74	30.86	64	6.6	28.99	196	1.11
VGG16	134.26	30.93	256	12.5	8.19	338	0.66

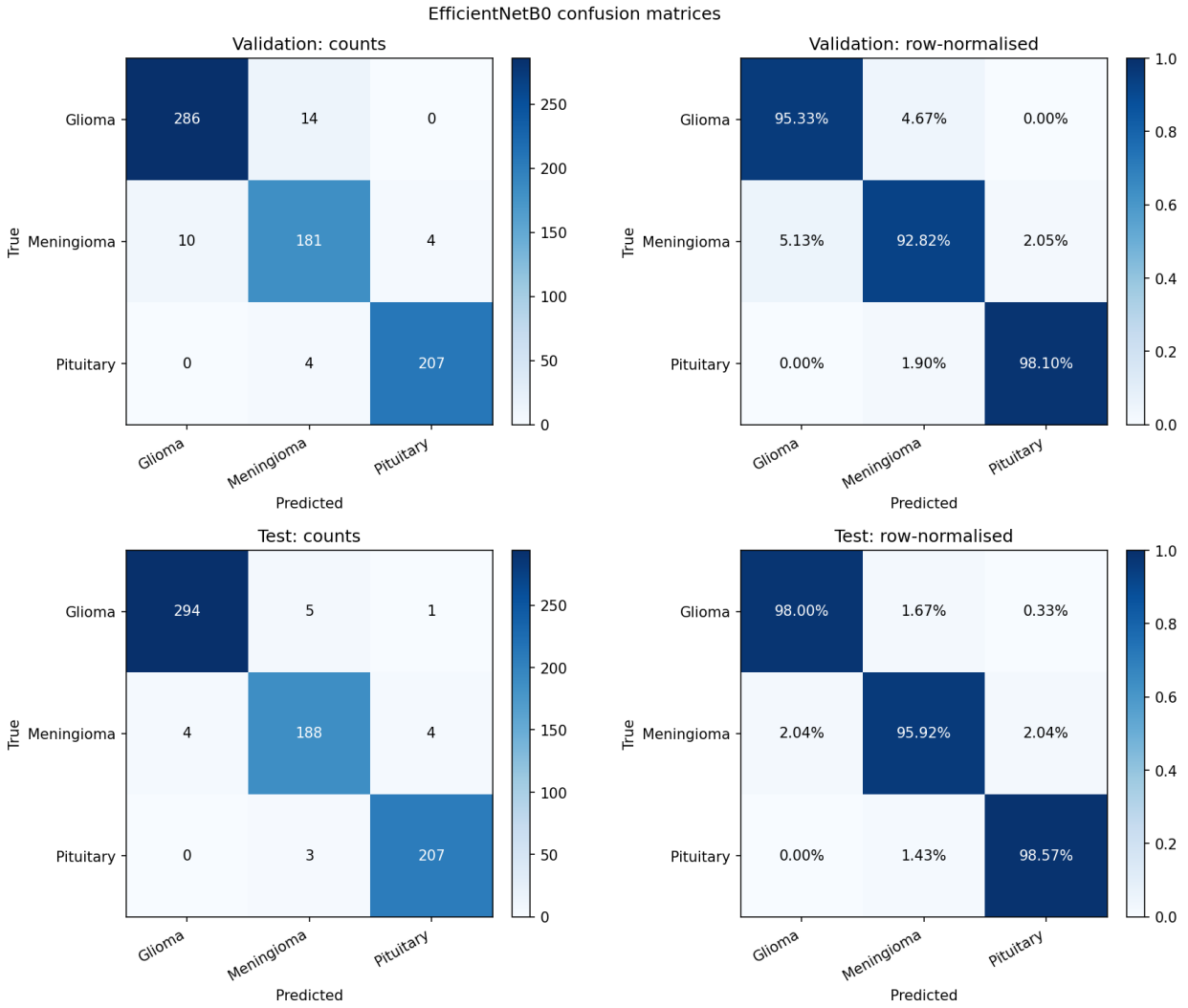


Figure 17: Confusion matrices for **EfficientNetB0** on the multiclass classification task, shown for the validation and held-out test splits. Most residual confusion occurs between meningioma and the other two classes.

The corresponding measurements for the multiclass runs are given in Table 11. The ordering by parameter count and GFLOPs is necessarily identical, since the same backbones are used, but the wall-clock and throughput figures differ because batch sizes were re-tuned for the larger multiclass dataset and because the training set is 45% larger. Two points stand out. First, EfficientNetB0 again dominates the accuracy-per-parameter measure at 24.33 points per million, more than twenty times the figure for either transformer. Second, single-image latency separates the two transformers sharply despite their near-identical size: Swin-Base requires 35.1 ms against 8.4 ms for ViT-B/16, because the shifted-window attention that gives Swin-Base its accuracy advantage also introduces sequential windowing operations that do not parallelise well at batch size one. Swin-Base therefore buys its leading accuracy at roughly four times the per-request latency of ViT-B/16 and more than twice that of EfficientNetB0.

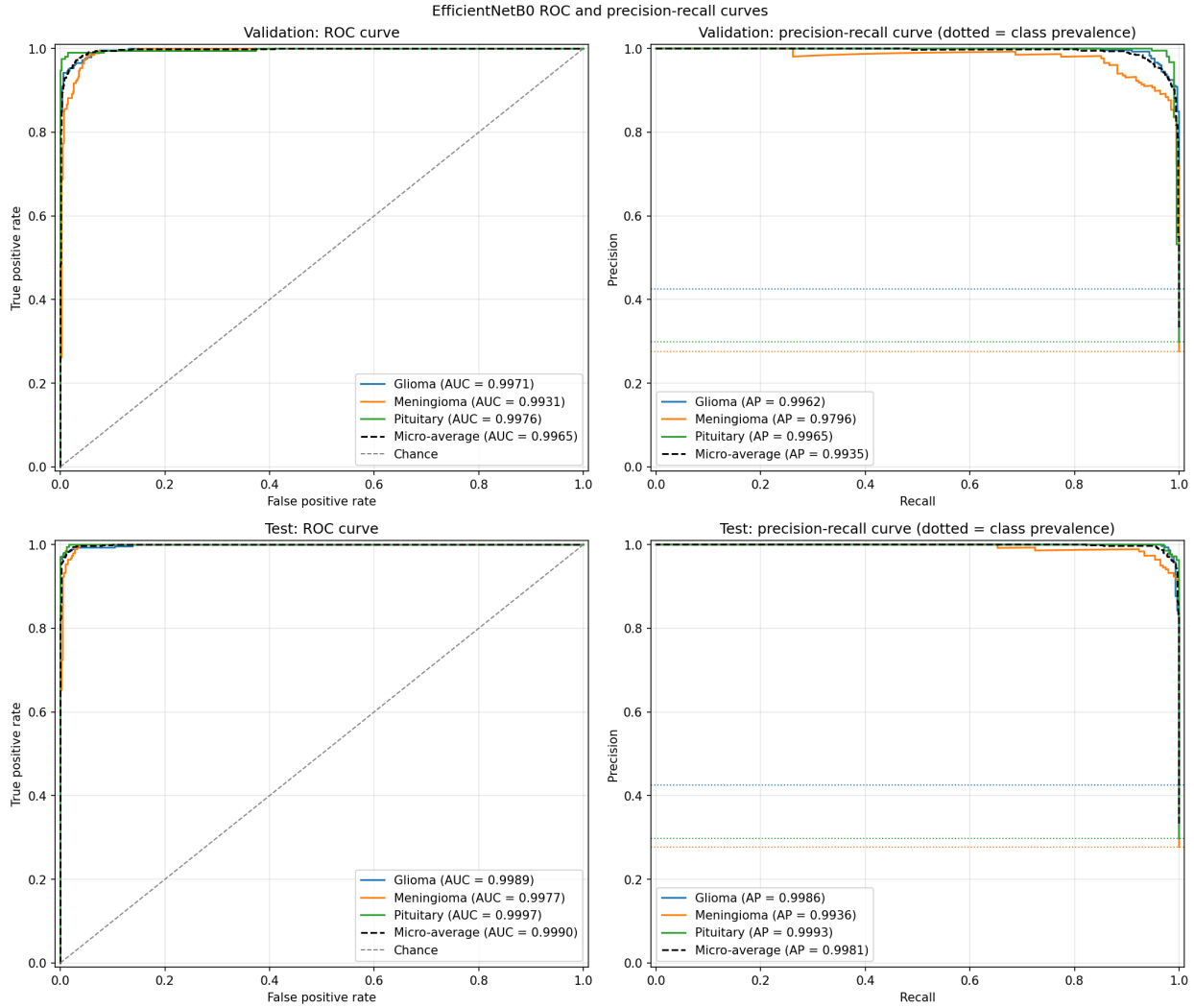


Figure 18: One-versus-rest receiver operating characteristic and precision–recall curves for **EfficientNetB0** on the multiclass test split, corresponding to a macro-averaged AUC of 0.9988.

Table 11: Computational efficiency comparison for the multiclass classification task, measured on dual NVIDIA Tesla T4 graphics processing units. Latency is the mean over single-image inference; "Thr." is throughput in images per second, measured at batch size 32, and "Acc./M" is test accuracy in percentage points per million parameters. Rows are ordered by parameter count.

Model	Params (M)	GFLOPs	Batch	Train (min)	Latency (ms)	Thr. (img/s)	Acc./M
EfficientNetB0	4.01	0.77	256	26.9	13.85	1471	24.33
ResNet50	23.51	8.17	256	8.1	7.82	792	4.15
InceptionV3	25.12	11.42	128	12.2	14.15	533	3.91
CoAtNet-0	26.67	8.42	128	28.0	13.23	499	3.69
ViT-B/16	85.80	22.54	32	7.4	8.39	290	1.05
Swin-Base	86.75	30.86	64	9.5	35.13	202	1.14

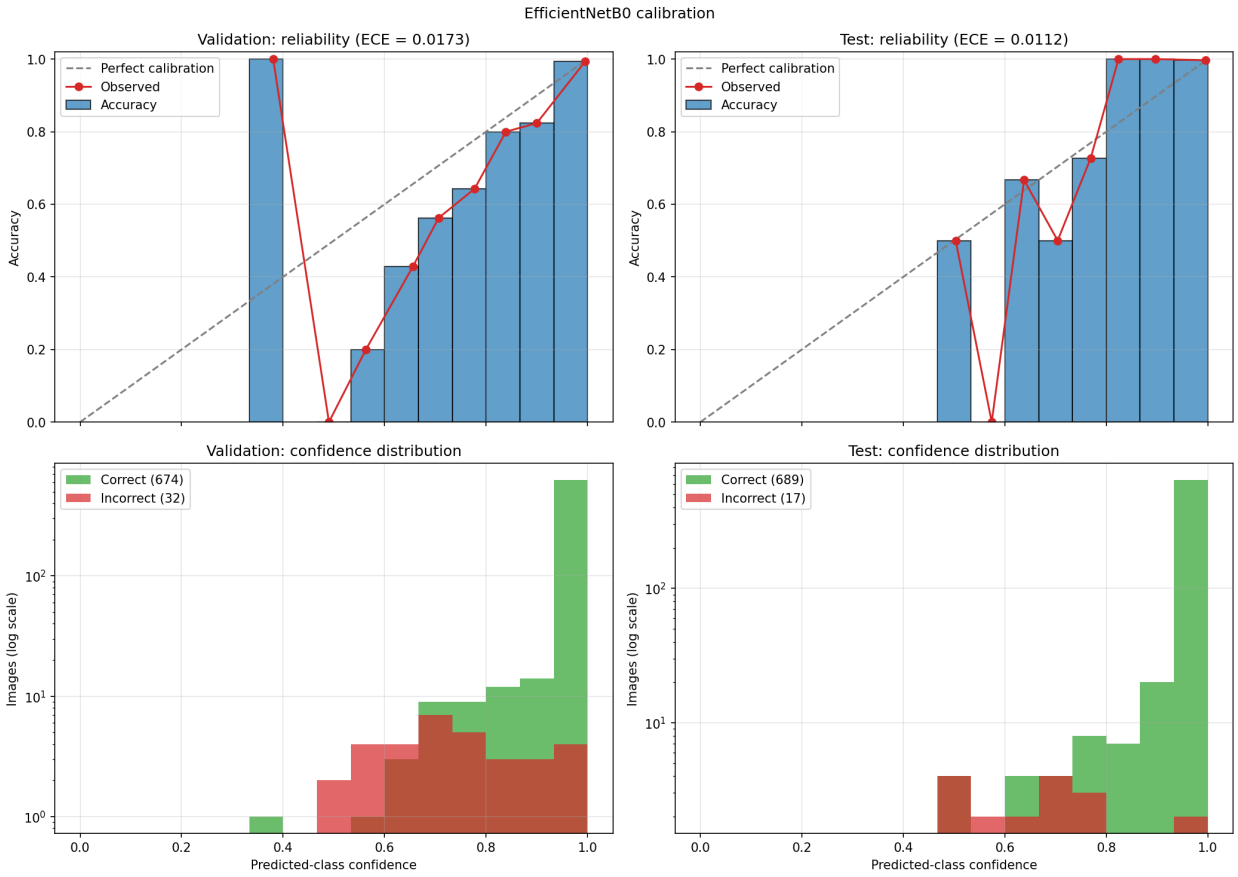


Figure 19: Reliability diagram for **EfficientNetB0** on the multiclass test split, corresponding to an expected calibration error of 0.0112.

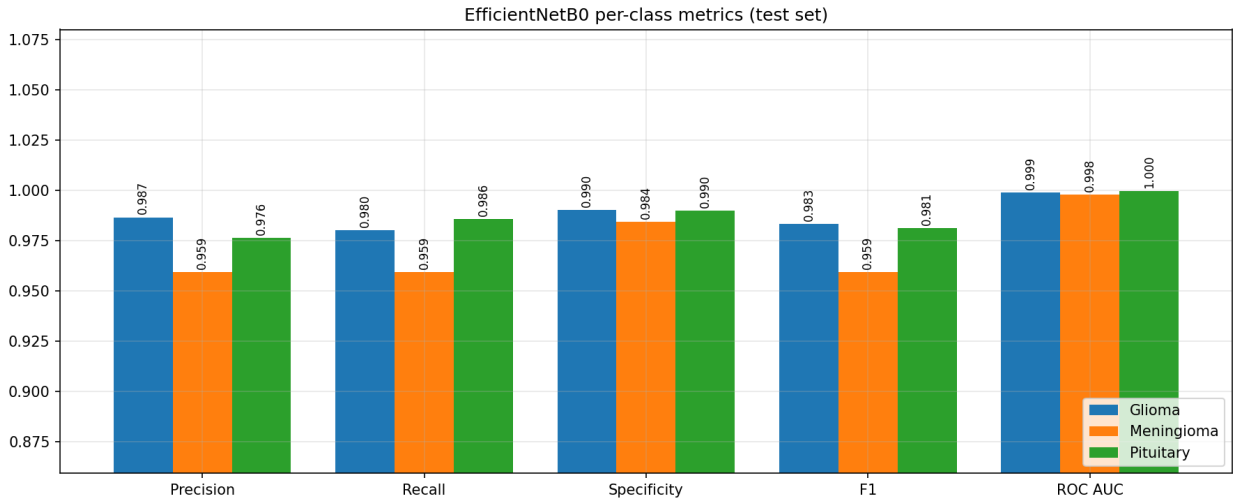


Figure 20: Per-class precision, recall, and F1 scores for **EfficientNetB0** on the multiclass test split, showing meningioma as the weakest of the three classes.

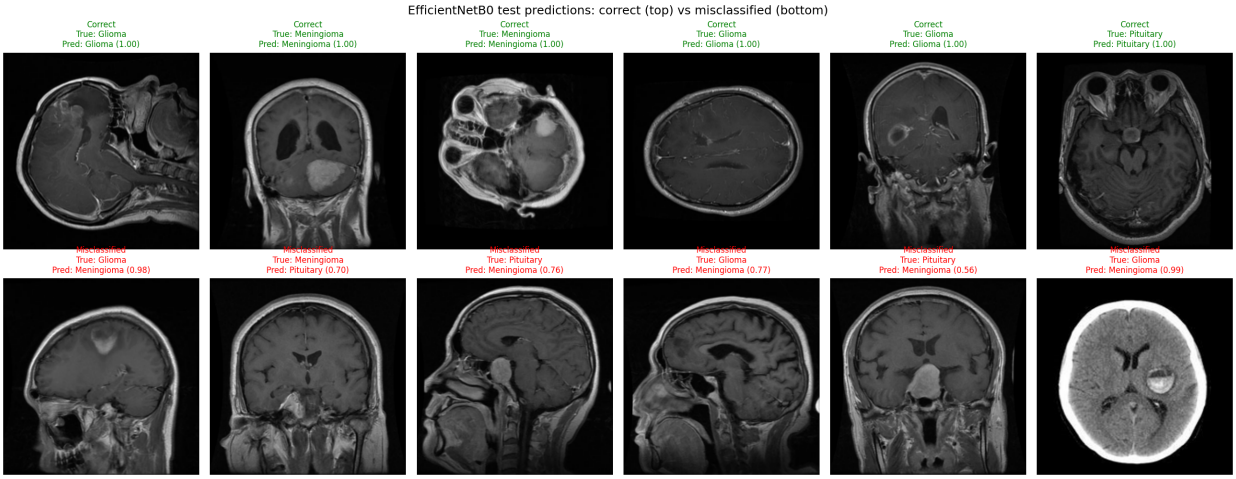


Figure 21: Representative test-set predictions of **EfficientNetB0** on the multiclass classification task, with the predicted tumour subtype and associated probability shown for each image.

Calibration quality, reported in Table 12, adds a dimension that accuracy alone does not capture. The four leading binary models are all well calibrated, with expected calibration errors between 0.0044 and 0.0084, meaning that a predicted confidence of 95% corresponds closely to a 95% empirical success rate. MobileNetV3-Large and VGG16, by contrast, show calibration errors of 0.0685 and 0.0437, an order of magnitude worse. This matters for deployment: a model whose confidence scores are unreliable cannot safely be used to triage cases by confidence or to defer uncertain cases to a human reader, even if its ranking performance measured by AUC appears strong. On the multiclass task all six models are reasonably calibrated, with Swin-Base best at 0.0076 and ViT-B/16 worst at 0.0266.

Table 12: Expected calibration error (ECE), Matthews correlation coefficient (MCC), and Cohen's kappa on the held-out test split for both tasks. Lower ECE indicates better agreement between predicted confidence and observed accuracy.

Task	Model	ECE	MCC	Kappa
Binary	DenseNet121	0.0044	0.9878	0.9877
	EfficientNetB0	0.0049	0.9878	0.9877
	ResNet50	0.0057	0.9878	0.9877
	InceptionV3	0.0078	0.9755	0.9754
	CoAtNet-0	0.0084	0.9878	0.9877
	Swin-Base	0.0235	0.9192	0.9181
	ViT-B/16	0.0323	0.8978	0.8975
	VGG16	0.0437	0.7673	0.7665
	MobileNetV3-Large	0.0685	0.8302	0.8161
Multiclass	Swin-Base	0.0076	0.9784	0.9783
	InceptionV3	0.0077	0.9741	0.9740
	CoAtNet-0	0.0084	0.9762	0.9761
	EfficientNetB0	0.0112	0.9632	0.9632
	ResNet50	0.0116	0.9633	0.9632
	ViT-B/16	0.0266	0.8550	0.8550

4.4. Interpretability Results

A performance figure alone does not establish that a model is attending to clinically relevant image regions rather than to incidental artefacts such as skull outline, field of view, or scanner-specific intensity patterns. To examine this, gradient-weighted class activation mapping was applied to the final convolutional stage of EfficientNetB0 for both tasks, producing heat maps that indicate which spatial regions most influenced the predicted class. Figure 22 shows the resulting maps for correctly classified binary detection cases, and Figure 23 shows the corresponding maps for correctly classified multiclass cases.

For correctly classified tumour-positive cases, the activation consistently concentrates on the lesion itself rather than on peripheral anatomy, which supports the interpretation that the model has learned a tumour-relevant representation rather than a dataset-specific shortcut. In the multiclass setting the attended regions are broadly consistent with the anatomical expectations for each subtype, with pituitary cases showing focused activation in the sellar region and glioma cases showing more diffuse activation across the affected hemisphere.

Figure 24 shows the equivalent maps for Swin-Base, computed at the final normalisation layer of its last attention stage, and permits a comparison between a convolutional and a hierarchical transformer backbone on the same task. The comparison is not reassuring. For meningioma the Swin-Base heat maps sit on the visible lesion, but for glioma and pituitary they largely highlight image corners and borders lying outside the brain entirely. Since Swin-Base is the most accurate subtype classifier in this study, this is a material finding: it suggests that part of its accuracy may derive from background or acquisition cues that correlate with class rather than from tumour morphology. Such cues are plausible here because the two source collections differ in native image size and framing, as Table 2 shows, and because class proportions differ between them, so border artefacts could carry genuine label information without carrying any clinical meaning.

This observation is qualitative and drawn from a single figure; the notebooks do not test it, and no measurement against annotated masks was performed. It nevertheless tempers the headline ranking. A model that attains 98.58% accuracy partly by reading image borders would not retain that accuracy on data acquired under a different protocol, and the comparison illustrates why interpretability evidence belongs alongside accuracy rather than after it. The activation maps of convolutional and attention stages are in any case not strictly commensurable, so the contrast with EfficientNetB0 should be read as a prompt for further investigation rather than as a measured difference.

Figures 25 and 26 show the maps for misclassified cases. The selection rule matters for their interpretation: these panels show the *most confident* errors, up to six per model, ranked by the probability assigned to the wrong class. They are therefore the failures in which the model was most assured, not representative or borderline ones. For EfficientNetB0 the binary panel exhausts the available errors, since only three test images were misclassified, whereas the multiclass panel shows six of seventeen. In these high-confidence failures the activation is not noticeably less focused than in the correct cases, which is the more troubling observation: a confidently wrong prediction supported by a plausible-looking activation map cannot be filtered out either by thresholding on confidence or by visual inspection of the map.

This should be read as a qualitative observation rather than a quantitative interpretability evaluation. No assessment against radiologist-annotated tumour masks was carried out, because the datasets used provide class labels only and contain no segmentation masks against which the attention maps could be scored. Quantifying attention overlap against expert annotation is identified in Section 5 as necessary future work.

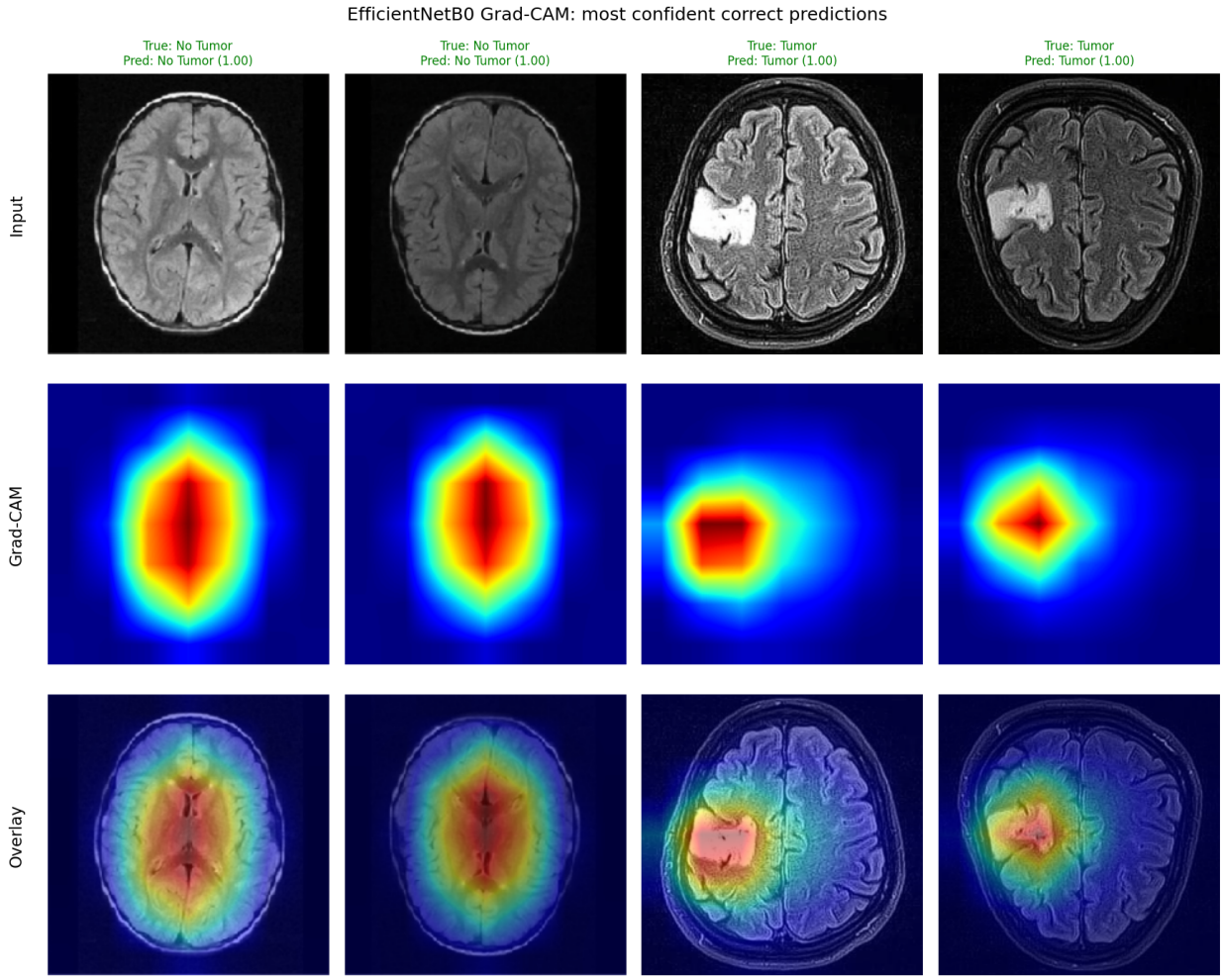


Figure 22: Gradient-weighted class activation maps for **EfficientNetB0** on correctly classified binary detection test cases, showing the two most confident correct predictions per class. Activation concentrates on the lesion rather than on peripheral anatomy.

4.5. Comparison with Existing Literature

Table 13 compares the present results with previously published studies using the same architectures. The ResNet50 and InceptionV3 results obtained here are consistent with, or exceed, the published benchmarks. Two discrepancies deserve comment. The VGG16 result of 88.32% falls well short of the 98.69% reported by Khan et al. [18]; this is attributable to differences in preprocessing, augmentation strategy, and train-test partitioning, and in particular to the fact that the present protocol fixes a single shared set of hyperparameters across all nine models rather than tuning each one individually, a choice that penalises architectures such as VGG16 that are sensitive to learning rate. The DenseNet121 result of 99.39% is marginally below the 99.51% reported by Noreen et al. [19], a difference of less than one image on a test set of this size and therefore not meaningful.

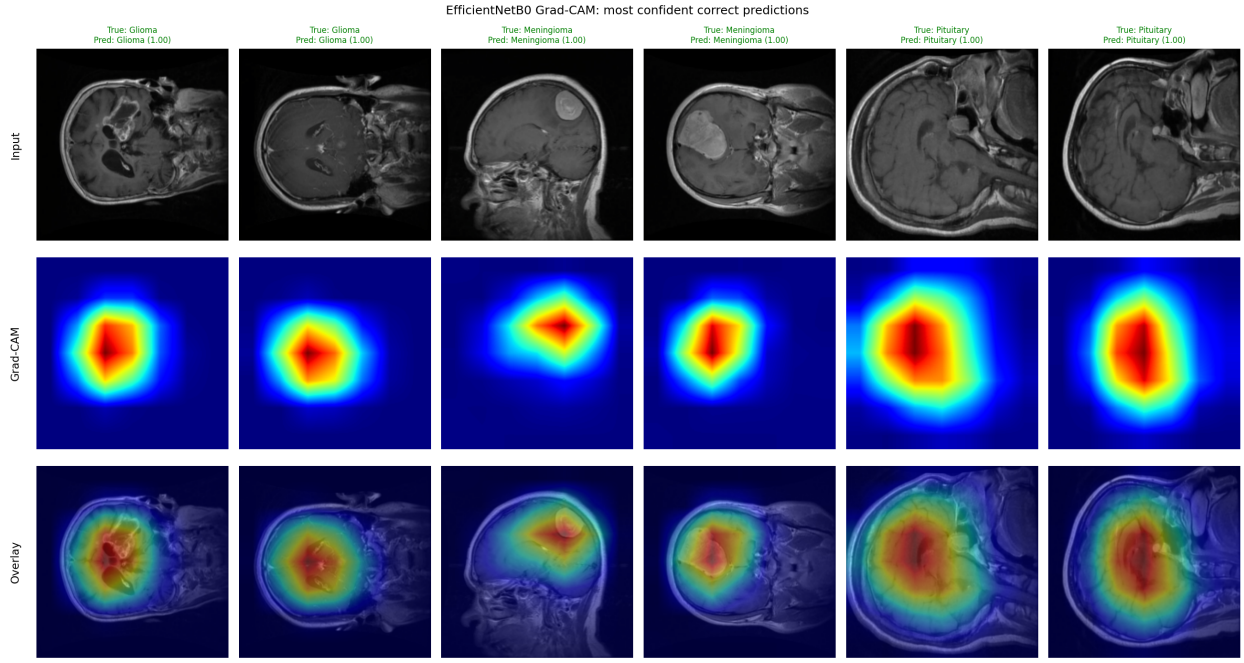


Figure 23: Gradient-weighted class activation maps for **EfficientNetB0** on correctly classified multiclass test cases, showing the two most confident correct predictions for each of glioma, meningioma, and pituitary.

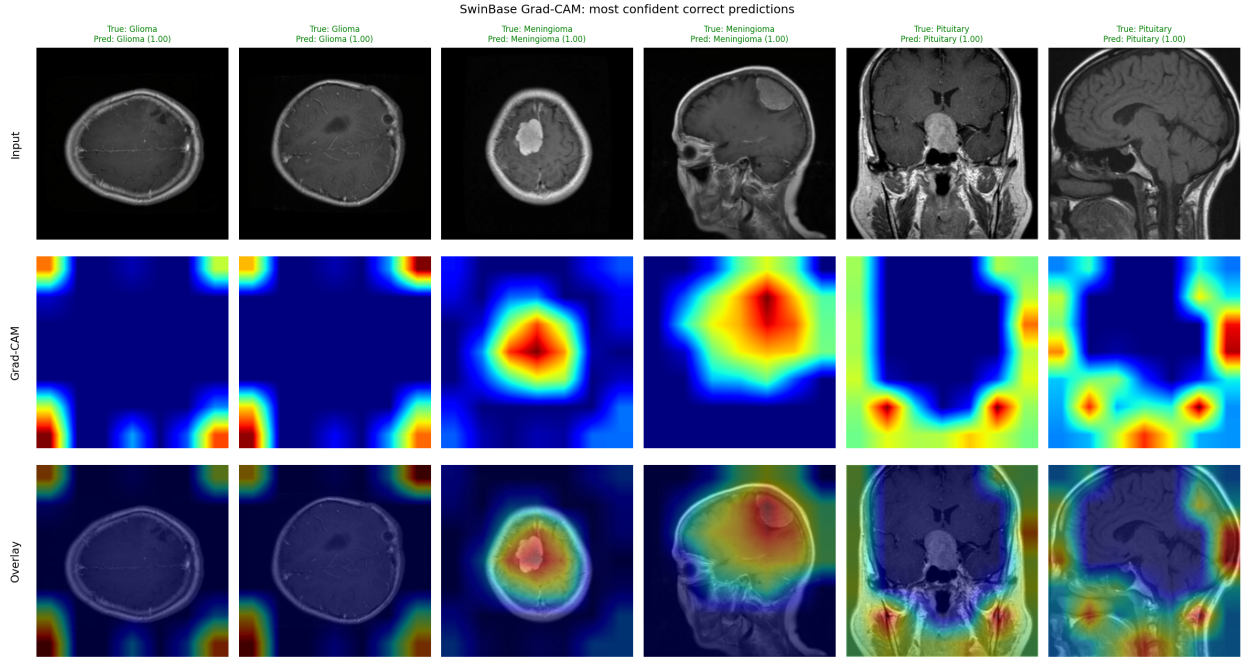


Figure 24: Gradient-weighted class activation maps for **Swin-Base** on correctly classified multiclass test cases, showing the two most confident correct predictions per class. Activation sits on the lesion for meningioma but falls largely on image corners and borders outside the brain for glioma and pituitary, suggesting partial reliance on non-anatomical cues.

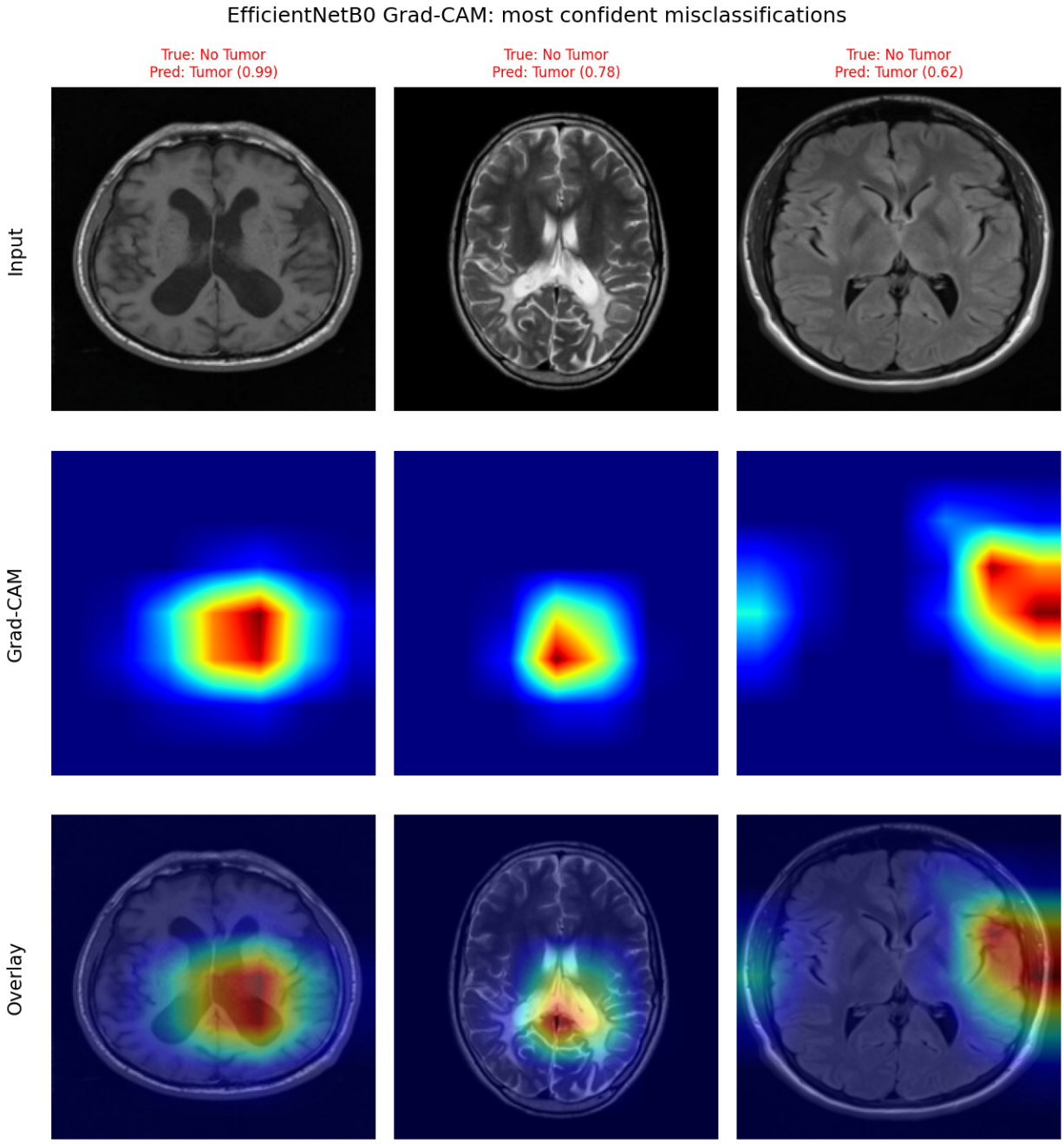


Figure 25: Gradient-weighted class activation maps for **EfficientNetB0** on misclassified binary detection test cases, ranked by confidence in the wrong class. Only three test images were misclassified, so all are shown.

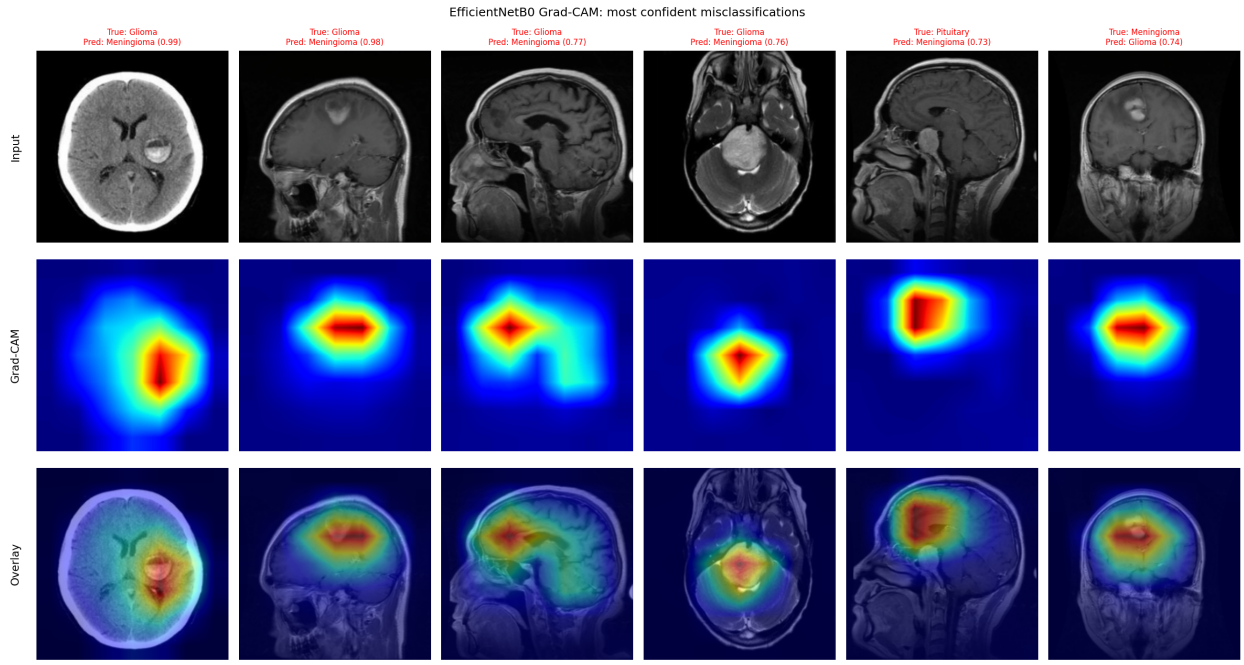


Figure 26: Gradient-weighted class activation maps for **EfficientNetB0** on the six most confident misclassified multiclass test cases, of seventeen errors in total. These are the failures in which the model was most assured, not representative ones.

More generally, direct comparison with published figures should be treated with caution, for three reasons. First, most of the studies in Table 13 evaluate on the 3,064-image Cheng collection without deduplication, whereas the multiclass results reported here are computed after removing 19.3% of the images as near-duplicates; deduplication removes exactly the images most likely to have inflated a reported score, so a slightly lower number obtained on a deduplicated split is not evidence of a weaker model. Second, as noted in Section 3.6, the multiclass pool here merges the original training and testing folders of the Bhuvaji dataset before drawing fresh stratified splits, so the test set used in this work is not the test set used by studies that adopt that dataset's official partition. Third, the models compared here share a single hyperparameter configuration rather than being individually tuned. The comparison is therefore included for orientation rather than as a like-for-like benchmark, and the absolute figures in the final column should not be cited as head-to-head wins or losses against the referenced studies.

Table 13: Comparison of the present results with previously published studies. Results obtained in this work are measured on the held-out test split.

Study	Model	Dataset size	Task	Reported accuracy	This work
Talo et al. (2019) [13]	ResNet50	253	Binary	95.23%	99.39%
Khan et al. (2020) [18]	VGG16	3,064	Binary	98.69%	88.32%
Noreen et al. (2020) [19]	DenseNet121	3,064	Binary	99.51%	99.39%
Rehman et al. (2020) [20]	InceptionV3	3,064	Binary	99.34%	98.77%
Deepak and Ameer (2019) [12]	InceptionV3	3,064	Multiclass	97.50%	98.30%

Several architectural innovations contribute to improved performance in medical imaging tasks. Residual learning in ResNet enables the training of deeper networks that capture hierarchical representations from low-level edges to high-level semantic features. Dense connections maximise information flow and feature reuse with comparatively few parameters, which explains why DenseNet121 matches the accuracy of ResNet50 with less than a third of its parameters. Inception modules capture tumours of varying size through parallel convolutional paths of different kernel sizes. Attention mechanisms in CoAtNet-0 and Swin-Base model long-range dependencies that are relevant for relating a tumour region to surrounding healthy tissue. Neural architecture search in EfficientNetB0 and MobileNetV3-Large optimises architectural decisions beyond human intuition, though the MobileNetV3-Large result shows that architectural efficiency does not by itself guarantee stable optimisation under a fixed shared training budget.

The clearest cross-cutting finding is what may be termed the transformer paradox: pure transformers underperformed while hybrid convolution-attention models excelled. ViT-B/16 was the weakest model on both tasks despite being among the largest, whereas CoAtNet-0 was among the strongest on both. With training sets of 2,277 and 3,293 images, there is insufficient data for a pure transformer to learn the spatial inductive biases that convolutional architectures encode by construction. The practical implication for medical imaging at this data scale is that attention should be adopted in hybrid form rather than in isolation.

Regarding model selection, the appropriate choice depends on the deployment constraint. Where accuracy is the primary consideration and computational resources are not limiting, Swin-Base is the strongest subtype classifier at 98.58% accuracy, and CoAtNet-0 offers almost the same accuracy at 98.44% for less than a third of the parameters and a threefold higher throughput. Where deployment is on constrained hardware, EfficientNetB0 is the clear recommendation: it reaches 99.39% detection accuracy and 97.59% subtype accuracy with 4.01 million parameters, 0.77 GFLOPs per image, and well-calibrated confidence scores. Where single-request latency dominates, ResNet50 provides the fastest single-image inference at 7.35 ms while retaining 99.39% detection accuracy, and it remains a reasonable default baseline for further research. No single model dominates on every axis, and the evidence presented here is intended to support a selection made against explicit deployment requirements.

5. Conclusion

This study benchmarked nine deep learning architectures for brain tumour detection and subtype classification under a single standardised protocol comprising identical stratified splits, identical preprocessing, shared hyperparameters, and a common evaluation suite, thereby addressing the absence of controlled cross-architecture comparison that has limited inter-study comparability in this field; perceptual hash deduplication removed 19.3% of the multiclass images before splitting, preventing near-duplicate leakage and ensuring that the reported figures reflect genuine generalisation. On binary detection, CoAtNet-0, DenseNet121, EfficientNetB0, and ResNet50 each reached 99.39% test accuracy, with CoAtNet-0 and EfficientNetB0 achieving perfect sensitivity and DenseNet121 and ResNet50 achieving perfect specificity and an AUC of 1.0000, while MobileNetV3-Large at 90.78% and VGG16 at 88.32% proved inadequate; on subtype classification, Swin-Base led at 98.58%, followed by CoAtNet-0 at 98.44% and InceptionV3 at 98.30%, with EfficientNetB0 and ResNet50 tied at 97.59% and ViT-B/16 trailing at 90.51%. Three findings carry beyond the leaderboard: the hybrid convolution-attention design of CoAtNet-0 consistently outperformed the pure transformer ViT-B/16 of triple the size, indicating that at this data scale attention must be paired with a convolutional prior; parameter efficiency and accuracy are not in tension, since EfficientNetB0 matched or approached far larger models at 4.01 million parameters, 0.77 GFLOPs per image, and 24.79 accuracy points per million parameters; and headline accuracy conceals clinically material differences in error profile and probability calibration, as models with near-identical accuracy differed in whether their errors were missed tumours or false alarms, and models with strong AUC such as MobileNetV3-Large proved badly calibrated. The principal limitations are that the datasets, at 3,253 and 4,705 images drawn from public two-dimensional slice collections, may not represent the heterogeneity of clinical practice across scanners, field strengths, and patient populations; that no external validation on institutional data was performed; that training was fixed at ten epochs with a single shared hyperparameter configuration and a single run per model, without cross-validation, per-model tuning, or ensembling, so that the leading models are statistically indistinguishable from one another; that batch size varied from 32 to 256 across models, leaving the optimisation budget uncontrolled and confounding architecture with number of gradient updates; that deduplication was applied to the multiclass track only, and there by an order-dependent procedure that silently resolves cross-class near-duplicates in favour of glioma; that Swin-Base was sourced from a different library and pretrained checkpoint in each of the two tasks, so its results are not comparable across them; that the Grad-CAM evidence indicates the best-performing subtype model attends partly to regions outside the brain, so its accuracy may rest in part on acquisition artefacts rather than pathology; that single two-dimensional slices rather than three-dimensional volumes or multi-sequence inputs were used; and that no integration with DICOM-compliant clinical workflows was attempted. Future work should therefore extend deduplication to the binary track and make it label-aware, equalise the optimisation budget across architectures by fixing the number of gradient updates rather than the number of epochs, incorporate additional public and institutional datasets including BraTS and TCGA, adopt k-fold cross-validation with multiple seeds, per-model hyperparameter tuning, and ensembling of the strongest architectures, extend the label space to tumour grade and localisation, move to three-dimensional and multi-sequence inputs, evaluate robustness on degraded and out-of-distribution images, quantify interpretability against radiologist-annotated masks rather than qualitatively, and develop DICOM-compliant inference pipelines with uncertainty estimation for prospective clinical assessment. With detection accuracy above 99% and subtype accuracy above 98% under leakage-controlled evaluation, the architectures examined here are accurate enough to warrant investigation as a second-reader aid rather than as an autonomous diagnostic, though the interpretability evidence indicates that accuracy alone does not establish that a model has learned pathology rather than acquisition artefact, and external validation is a precondition for any clinical claim; the accompanying open code and per-model artefacts are released so that both the results and the caveats attached to them can be reproduced and extended.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets used in this study are publicly available and are cited in Section 3.2. The complete code, trained model checkpoints, and per-model evaluation artefacts are available at the repository given in Appendix C.

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Appendix A. Complete Binary Detection Results

Table A.1 reports the training and validation metrics at the best epoch for each model on the binary detection task, where the best epoch is the one giving the highest validation AUC. These are validation-split figures; the corresponding test-split results are given in Table 6. Figure A.1 shows the resulting stratified split distribution and Figure A.2 the native image properties of this dataset.

Table A.1: Binary detection training and validation metrics at the best epoch, selected by validation AUC.

Model	Epoch	Train loss	Train acc.	Train AUC	Val. loss	Val. acc.	Val. AUC
CoAtNet-0	6	0.0114	0.9969	0.9999	0.0322	0.9898	0.9999
DenseNet121	4	0.0099	0.9969	1.0000	0.0042	1.0000	1.0000
EfficientNetB0	4	0.0307	0.9937	0.9996	0.0178	0.9959	0.9999
ResNet50	7	0.0042	0.9990	1.0000	0.0184	0.9939	0.9999
InceptionV3	3	0.0314	0.9922	0.9997	0.0106	0.9959	1.0000
Swin-Base	6	0.0953	0.9665	0.9938	0.0928	0.9693	0.9973
ViT-B/16	10	0.1649	0.9439	0.9837	0.1449	0.9406	0.9899
MobileNetV3-Large	10	0.0023	0.9990	1.0000	0.3995	0.8811	0.9993
VGG16	10	0.2590	0.8872	0.9603	0.2421	0.8975	0.9687

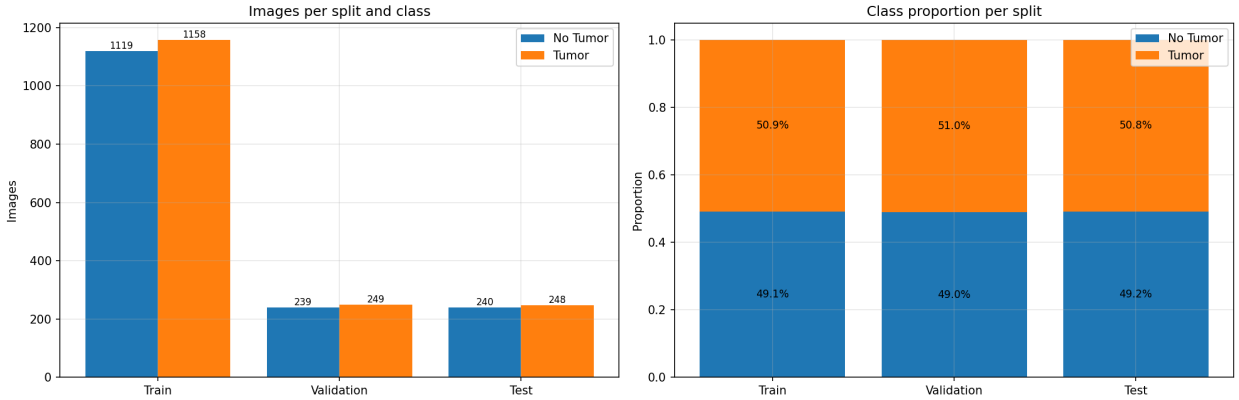


Figure A.1: Stratified training, validation, and test split distribution for the binary detection dataset.

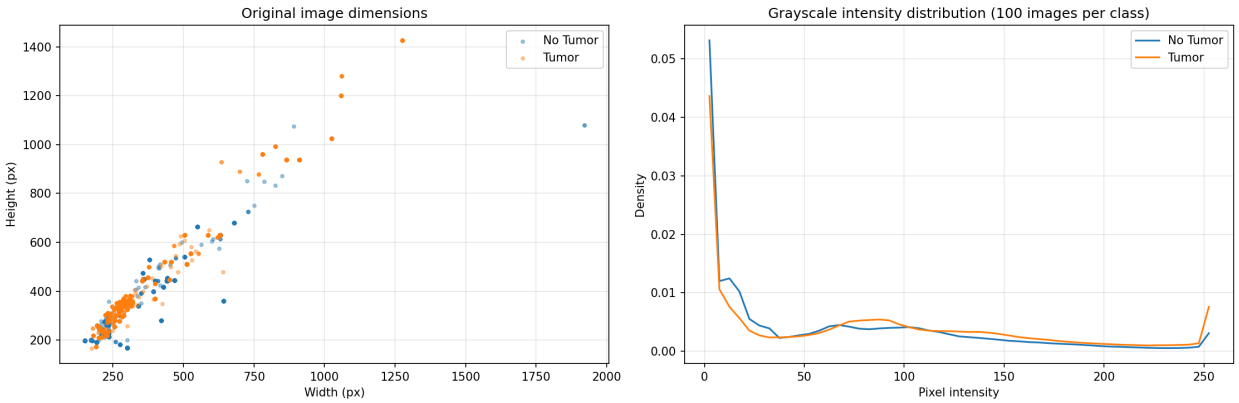


Figure A.2: Native image dimension and intensity distributions for the binary detection dataset, showing the wide variation in resolution and field of view across the two contributing sources. Summary statistics are given in Table 2.

Appendix B. Complete Multiclass Results

Table B.1 reports the training and validation metrics at the checkpointed epoch for each model on the multiclass task, where that epoch is the one giving the highest validation macro one-versus-rest AUC. Because this metric saturates near 0.997, it distinguishes only weakly between late epochs, and for several models validation accuracy continued to rise after the checkpointed epoch; Swin-Base, for example, is checkpointed at epoch 7 with a validation accuracy of 97.31%, although its validation accuracy reaches 98.16% at epochs 9 and 10. These are validation-split figures; the corresponding test-split results are given in Table 8. Figure B.1 shows the post-deduplication class distribution, Figure B.2 the resulting stratified split distribution, and Figure B.3 the native image properties of this dataset.

Table B.1: Multiclass classification training and validation metrics at the checkpointed epoch, selected by validation macro one-versus-rest AUC.

Model	Epoch	Train loss	Train acc.	Train AUC	Val. loss	Val. acc.	Val. AUC
Swin-Base	7	0.0333	0.9884	0.9996	0.0861	0.9731	0.9974
CoAtNet-0	9	0.0246	0.9928	0.9998	0.1019	0.9660	0.9975
InceptionV3	8	0.0132	0.9966	1.0000	0.1061	0.9745	0.9963
EfficientNetB0	9	0.0396	0.9893	0.9997	0.1069	0.9547	0.9959
ResNet50	9	0.0121	0.9954	1.0000	0.0939	0.9731	0.9981
ViT-B/16	10	0.3040	0.8808	0.9712	0.3419	0.8810	0.9689

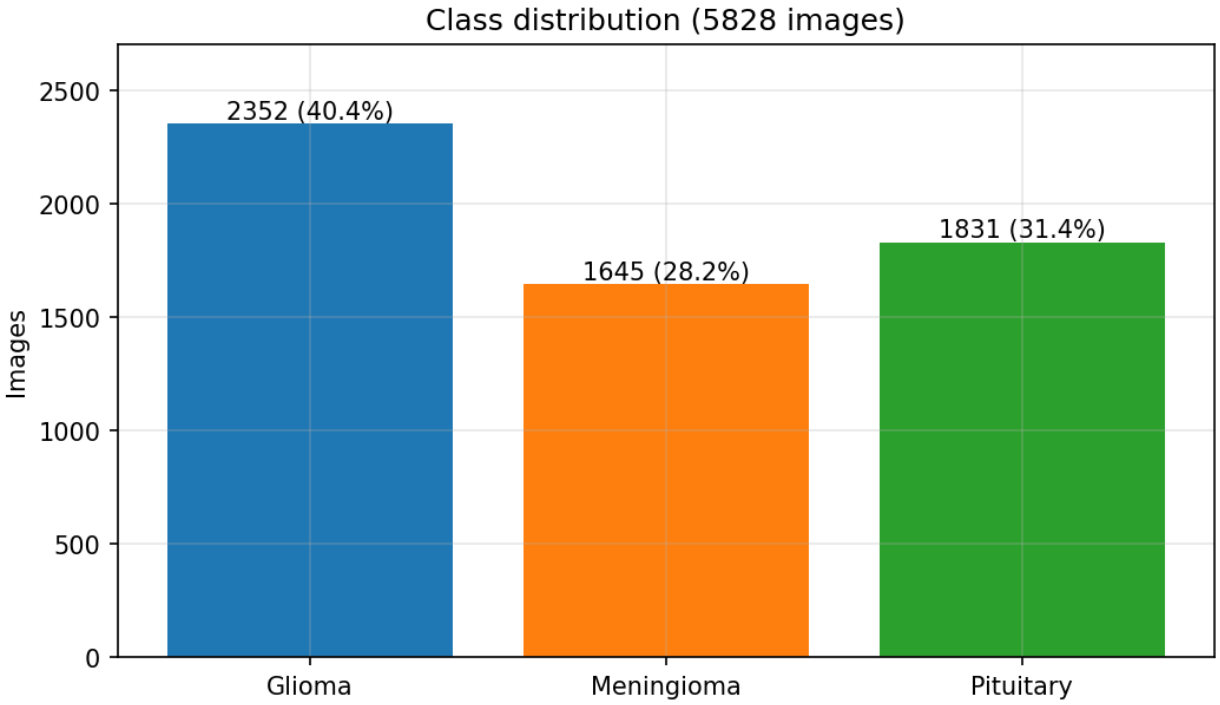


Figure B.1: Class distribution of the combined multiclass dataset *before* deduplication, covering all 5,828 images. The exploratory characterisation cells run on the raw pool; the post-deduplication counts used for training are given in Figure 5.

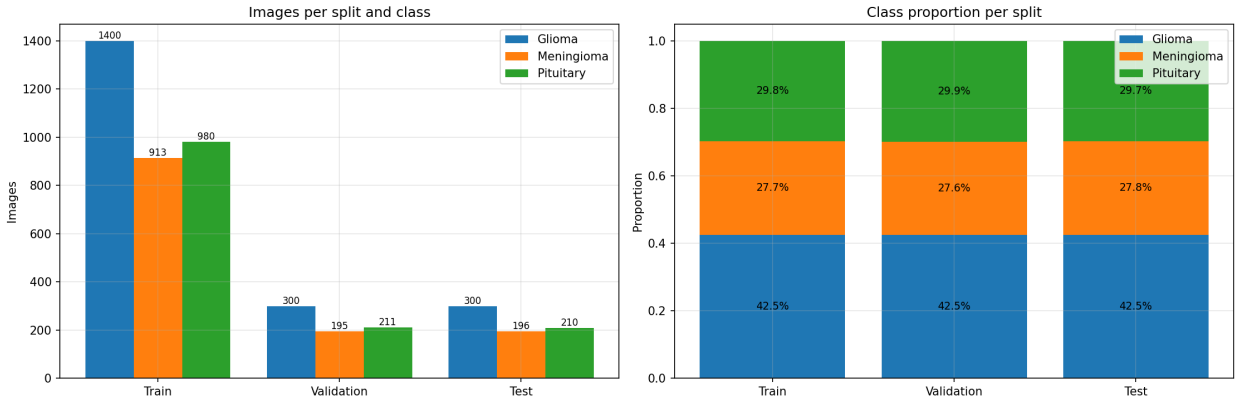


Figure B.2: Stratified training, validation, and test split distribution for the multiclass classification dataset.

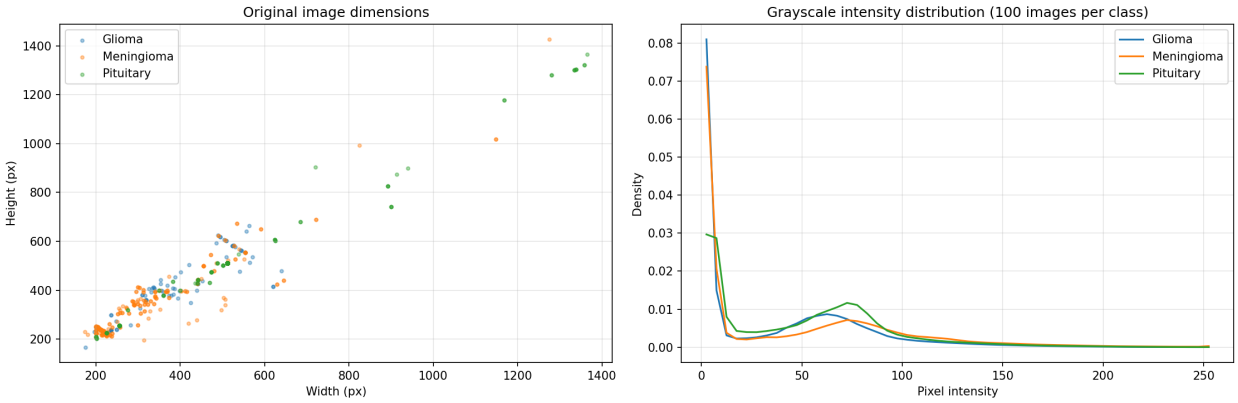


Figure B.3: Native image dimension and intensity distributions for the multiclass classification dataset before deduplication, which is markedly more homogeneous than the binary collection, with a median resolution of 512×512 pixels in every class. Summary statistics are given in Table 2.

Appendix C. Code Repository

The complete code for all experiments, including data preprocessing, perceptual hash deduplication, model training, evaluation, and figure generation, together with the trained checkpoints and the full per-model metric artefacts, is available at https://github.com/parthmital/NeuroScan-AI.